

Resource-backed loans and ecological efficiency of human development: Evidence from African countries

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ABSTRACT

The main challenge for most African countries is to find an alternative source of funding that can meet the needs of present generation without jeopardizing the needs of the future generations. Indeed, budgetary pressures are as defaults and economic recessions, combined with difficulties in accessing financial markets, which are often triggered by a series of sovereign debt restructurings and defaults, and have led some resource-rich African countries to turn to resource-backed loans to finance infrastructure projects to achieve long-term sustainable development goals. In this paper, we use the entropy balancing method to test whether resource-backed loans have a causal impact on the ecological efficiency of human development of African economies. Across a battery of robustness checks, the estimation results indicate that resource-backed loans have a positive and statistically significant impact on ecological efficiency of human development. Nonetheless, when resource-backed loans are disaggregated to run the regressions, we find that oil-backed loans and minerals resource-backed loans increase ecological efficiency of human development while cocoa and tobacco resource-backed loans are negatively associated to ecological efficiency of human development. The main findings of the study suggest that resource-backed loans agreements in general, and cocoa and tobacco in particular, should include strict clean development provisions and the implementation of actions to mitigate the negative environmental impact of these loans for borrowing countries.

1. Introduction

Resource-backed loans have emerged as an alternative financing tool for resource-rich developing countries that find it difficult to mobilize resources on the financial markets to finance their economic development due to their high financial risks (Coulibaly, 2023; Wang et al., 2023; Mihalyi et al., 2020, 2022).¹ To benefit from this type of loan, the debtor country must necessarily have natural resources. In fact, in the clauses of these loans, natural resources represent an asset that serves as collateral and repayment to the lenders (government, public enterprise, private sector and/or international financial institutions) can be made using the income stream from the natural resources or using the natural resources as payment in kind (Coulibaly, 2023; Mihalyi et al., 2020, 2022).

Latin American and sub-Saharan African countries are the main

signatories of these resource-backed loans, but African countries are the main beneficiaries of resource-backed loans.² By way of illustration, from 2004 to 2018, 30 of the 52 resource-backed loans identified during this period in these two regions were signed by African countries, while the remaining 22 were signed by Latin American countries (Coulibaly, 2023; Mihalyi et al., 2020, 2022). By financing large-scale projects, resource-backed loans can improve the well-being of the populations concerned and thus affect the ecological efficiency of human development. This is the case, for example, with the Sicomines agreement in the DRC, which lists 25 major projects (road, rail and health), and the various road and health projects in Guinea, as well as the construction of a university (Mihalyi et al., 2020, 2022). A great deal of funding has been provided in the energy sector. For example, the Bui dam in Ghana, the rural electrification programme in Zimbabwe and numerous hydroelectric dams. A road project in Southern Sudan and a housing

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¹ These are countries that have defaulted on their sovereign debt and have undergone a series of sovereign debt restructurings.

² These countries are: Angola, Brazil, Chad, Democratic Republic of Congo, Ecuador, Ghana, Guinea, Niger, Republic of Congo, Sao Tome and Principe, Sudan, South Sudan, Venezuela and Zimbabwe (see, Coulibaly, 2023; Coulibaly et al., 2022; Mihalyi et al., 2020).

project in Angola complete the list (see, for instance, [Mihalyi et al., 2020](#)). In fact, resource-backed loans, by increasing public investment, create certain benefits such as in facilitating trade (through road infrastructure), boosting GDP, increasing the share of educated, healthy and less poor population which are key pillars of Sustainable Development Goals. Consequently, resource-backed loans will end up affecting ecological efficiency and human development. In the same vein, it is possible that the financial resources from resource-backed loans can be used to finance renewable energy sources and green projects to improve the environment, and people's living conditions, which will have a positive impact on ecological efficiency and human development. In addition, resource-backed loans backed by minerals can accelerate the pace of minerals production which may likely increase the supply of minerals that are critical to produce batteries for energy transition. In the same vein, it is possible that resources from resource-backed loans could be used to finance environmental policies.

However, the collapse in commodity prices could increase the nominal amount of resource-backed loans already signed ([Mihalyi et al., 2020, 2022; Coulibaly, 2023](#)), as these are usually contracted in U.S. dollars and consequently cause borrowing countries to overexploit natural resources to repay the loans. As a result, resource-backed loans could affect the country's ecological efficiency by accelerating hydrocarbon production. Indeed, repayment of the resource-backed loans could lead the government either to begin exploiting oil resources or to accelerate the rate of production of minerals and hydrocarbons in order to meet the repayment schedule. This could accelerate the rate of depletion of natural resources and increase greenhouse gas emissions in the country, as fossil fuels are responsible for 73% of global greenhouse gas emissions ([UN, 2021](#)). This will lead to a deterioration in the ecological efficiency of human development. In the same vein, debt repayment is likely to lead to government budget cuts in environmental spending, adversely affecting environmental quality ([Gullison and Losos, 1993](#)).

Similarly, by accelerating the loss of forest cover, resource-backed loans could affect ecological efficiency through CO₂ and greenhouse gas emissions, as forests are known for their carbon sequestration. Resource-backed loans may also jeopardize the public debt situation ([Horn et al., 2021](#)), and prevent countries from mobilizing financial resources to fund sustainable development policies, as unsustainable public debt reduces fiscal space and weakens national capacity to respond to climate change and environmental problems combined with inequality, poverty and food insecurity. Given that resource-backed loans offer a wide range of public policies, they can have a negative impact on environmental sustainability, as public policies have a significant negative impact on deforestation ([Motel et al., 2009](#)). They can have a negative impact on ecological sustainability by intensifying the extraction activities of mining and oil companies. In fact, it has been shown that the activities of extractive industries cause great damage to the forest landscape ([Kinda and Thiombiano, 2021; Hiron, 2013](#)).

Theoretically, the environmental impact of resource-backed loans is somewhat ambiguous. Although there are a handful of studies on the impact of public debt on the environment ([Boly et al., 2022; Bese, 2021; Fodha and Seegmuller, 2014; Culas, 2006](#)), the literature is still silent on the impact of resource-backed loans on the ecological efficiency of human development, while the ecological and environmental impact of resource-backed loans might be different from the ecological and environmental impact of other types of public debt. In practice, the mechanism by which natural resource-backed loans affect ecological efficiency is different from those by which other non-resource-related public debt affects the ecological efficiency of human development. In fact, resource-backed loans put implicit pressure on the borrowing country to undertake fossil fuel production, an activity associated with significant ecological and environmental risks, whereas with other types of loans there are no constraints and no formal obligation on the government to exploit natural resources to repay the loans. Furthermore, due to the lack of transparency around resource-backed loans

([Coulibaly, 2023; Horn et al., 2021; Rivetti, 2021; Mihalyi et al., 2020, 2022](#)), previous studies on the impact of public debt on ecological efficiency of human development in developing countries may not take into account the ecological and environmental impact of resource-backed loans, as resource-backed loans may not be reflected in public debt statistics ([Coulibaly, 2023; Horn et al., 2021; Rivetti, 2021; Mihalyi et al., 2020, 2022](#)). Within this background, the handful of this study is to fill this gap in the economic literature by analyzing the impact of resource-backed loans on the human development index measured by the ecological efficiency of human development in African countries in order to enrich the political debate on climate change, public debt, and the exploitation of natural resources.

We took an interest in the human development index measured by the ecological efficiency of human development constructed by [Hickel \(2020\)](#). The purpose of using this index is that it integrates both human development and ecological impact. Unlike other development indices, such as the one developed by [Kovacevic et al. \(2010\)](#) and [Hickel \(2020\)](#)'s index illustrates how, and when it comes to development in the 21st century, poor nations are the «easy» part. This means that it is possible for poor countries to achieve high levels of human development with sustainable levels of ecological impact, and this for example can be done by investing in universal social goods.

The fact is that sub-Saharan African countries are now suffering disproportionately from the negative impacts of climate change and environmental degradation, with significant costs to economies and livelihood systems; climate change is now beginning to reverse key development indicators in some regions as agricultural yields decline and hunger rates increase ([Hickel, 2020](#)). In this regard, HDI embodies a contradiction in which the process of generating high levels of development in some countries curbs development or even leads to under development—in others. According to [Hickel \(2020\)](#), a development indicator that claims to be universal, such a contradiction is indefensible. In fact, the idea is that any ecologically rational vision of human development must respect the principle of planetary boundaries. In other words, resources must be mobilized to enhance human development, but without violating the parameters of ecological sustainability, as recommended by [Raworth \(2012, 2017\)](#). The aim is to achieve both human development and ecological sustainability. Given that the Sustainable Development Index (SDI) developed by [Hickel \(2020\)](#) addresses these concerns by focusing on the ecological efficiency of nations in achieving human development. This paper examines the link between this variable and resource-backed loans in African countries.

To identify the causal effect of resource-backed loans on ecological efficiency of human development, we rely on entropy balancing, an impact analysis method developed by [Hainmueller \(2012\)](#). Using a sample of 45 African countries over the period 1990–2019, we show that resource-backed loans adoption increases the ecological efficiency of human development in resource-backed loans countries, relative to non-resource-backed loans countries. Moreover, we emphasize that this finding is robust to wide set of specification, including a placebo test, adding several additional control variables, modifying the sample design, and using alternative estimation methods such as propensity score matching, Difference-in-Difference, panel fixed effects, IV-2SLS, and system-GMM estimator. Analysis of transmission channels shows that the positive effect of resource-backed loans on ecological efficiency of human development is mediated by education level, GNI per capita, expected years of schooling. The possible explanation for this result is that public investments (housing, roads, schools, hospitals, dams, etc.) made by RBLs will boost GDP and GNI, human capital through the level of education and the expected years of schooling, which in turn improve the sustainable development index measured by ecological efficiency of human development. However, heterogeneity tests show that the impact of resource-backed loans may depend on type of resource-backed loans, public debt level, EITI implementing, and certain structural factors in particular, government effectiveness, interest on debt, urban population, fiscal discipline, and natural resources abundant.

The rest of the paper is organized as follows. Section 2 reports some stylized facts. Section 3 reviews previous studies. Then, Section 4 describes the empirical methodology, while Section 5 presents the data and descriptive statistics. The main results are presented in Section 6, while in Section 7 and 8, we examine the sensitivity of these results and analyses the possible transmission channels. Section 9 explores heterogeneity tests. Finally, Section 10 concludes the study and presents the main policy recommendations.

2. Stylized facts

Before performing the main analysis, it is instructive to look at some stylized facts on the link between ecological efficiency of human development and resource-backed loans. To do this, we begin by analyzing some statistics on resource-backed loans in Africa. Thus, looking at the composition of the 52 loans backed by natural resources listed by Mihalyi et al., (2020, 2022), 43 were backed by oil, 6 by minerals, 2 by cocoa and 1 by tobacco. The total amount of resource-backed loans underwritten between 2004 and 2018 was valued at \$164 billion, of which \$98 billion went to Latin American countries and \$66 billion to sub-Saharan African countries. It should be noted that Guinea is Africa's largest recipient of resource-backed loans over the last few decades after Angola (see, Fig. 1a). In 2017, the country signed a \$20 billion resource-backed loans (almost 200% of the country's GDP) with a consortium of Chinese institutions. The Republic of Congo is the second largest recipient of resource-backed loans in Africa, having signed a resource-backed loan of \$1.6 billion in 2006, equivalent to 21% of the country's GDP that year. The Sicomines infrastructure agreement signed by the Democratic Republic of Congo with China in 2008 follows closely behind. The loan amounted to \$3 billion, equivalent to 16% of the country's GDP (Coulibaly, 2023; Mihalyi et al., 2020, 2022). The unconditional correlation between the two variables (see the Appendix, Table A11) is positive and significant at the 1% level. Next, we compare sustainability development index (SDI) measured by ecological efficiency of human development level in RBL and non-RBL countries, respectively; Fig. 2a shows that this ecological efficiency tends to be higher in RBL countries (on average around 0.49% for RBL countries versus around 0.47% for non-RBL countries). Moreover, based on the approach of Neuenkirch and Neumeier (2016), we compare the sustainability development index before and after the adoption of resource-backed loans in RBL countries with the sustainability development index in non-RBL countries; Fig. 2b shows that, despite lower sustainability development before the adoption of RBL, average sustainability development becomes higher after the adoption of resource-backed loans in RBL countries compared to non-RBL countries. Finally, we check how sustainability development index changes in the control countries using a before-after analysis of RBL adoption in both groups of countries.³ Fig. 2c shows that, although the sustainability index increases after RBL adoption in both groups, the increase is significant and more pronounced in RBL countries (with an increase of 15%, t -stat = 6.51 and p -value = 0.00) compared to non-RBL countries (with an increase of only 7%). In addition, a closer look at the Fig. 1b reveals a difference in the sustainable development index between RBL and non-RBL countries. Indeed, RBL countries exhibit a median (the middle line of the box), 25th (bottom hinge of the box), and 75th (top hinge of the box) percentiles of sustainable development index higher than non-RBL countries. This finding is justified by the fact that the median line, bottom hinge, and top hinge of the box for RBL countries are respectively higher those of non-RBL countries.

To conclude this section, when we graphically analyze the scale of resource-backed loans by comparing the level of the sustainable

development index (SDI) measured by the ecological efficiency of human development, we realize that countries that have signed a resource-backed loan have, in average, a higher level of ecological efficiency of human development than non-signatory countries (see, Fig. 1b; Fig. 2a; and Fig. 2c). However, the upward trend observed in both groups of countries indicates that it would be misleading to estimate the effect of resource-backed-loans on sustainable development index simply by comparing sustainable development index before and after RBL adoption, because this does not reflect a causal relationship. To avoid overestimating the effect of RBL, we use observations from non-RBL countries selected based on the matching procedure as a control group to estimate the counterfactual result.

3. Relevant literature

The literature on the economic, social, and environmental impacts of resource-backed loans in developing countries, particularly in Africa, is nascent and relatively few studies have been done (Coulibaly, 2023; Mihalyi et al., 2020, 2022; Coulibaly et al., 2022; Rivetti, 2021; Bräutigam and Gallagher, 2014). From this point of view, our work joins the few studies on sovereign debt, the environment, and sustainable development, as resource-backed loans are a special type of external debt. In subsection 3.1, we discuss the theoretical link, while subsection 3.2 explores the existing empirical link. Finally, in subsection 3.3, we discuss and formulate our various testable hypotheses.

3.1. Review of the theoretical framework

This sub-section analyzes the transmission mechanisms by which budgetary instruments, and sovereign debt in particular, can affect environmental sustainability and human development. We start the discussion on how sovereign debt affects the quality of the environment through economic growth. In fact, high public debt could harm economic growth (Eberhardt and Presbitero, 2015; Reinhart and Rogoff, 2010) and harm environmental quality by hampering the implementation of environmental protection policies. This could compromise the ability of indebted countries to engage in a sustainable energy transition over the long term. Besides, Van den Bergh (2013) argues that public debt can limit the ability to mobilize national resources and thus invest in climate change adaptation in the context of developing countries. In a similar vein, repaying the debt burden may force highly indebted economies to increase pressure on the natural environment in order to boost public revenues (Coulibaly, 2023; Combes et al., 2015), leading to heavy deforestation and depletion of natural resources (Coulibaly, 2023; Kinda and Thiombiano, 2021). Similarly, a recent study by Osakede and Adeleke (2022) showed that public borrowing negatively affects human development and education in several African countries. Akam et al. (2022) studied the effects of sovereign debt on the environment in just four African countries and found that increasing sovereign debt does not contribute to environmental sustainability, but rather degrades the environment.

The overlapping generations model of environmental endogenous growth (OLG) could also be mobilized to explain the mechanism of the relationship between public debt and environmental policy. This model predicts that production deteriorates the quality of the environment, thus harming the well-being of future generations. Public spending to reduce pollution is financed by taxes and public debt, but the government is constrained not to increase the ratio of public debt to production (Fodha et al., 2018; Fodha and Seegmuller, 2014). According to these authors, there are two stable states with opposite properties: in the stable state, higher debt reduces capital but improves environmental quality, while opposite effects are observed in the unstable state. Boly et al. (2022) also used this same model but with different configurations than those used by Fodha et al. (2018) and Fodha and Seegmuller (2014). In the configuration of Boly et al. (2022), two major aspects are incorporated. Firstly, they relax the classic balanced budget assumption

³ Given that 2004 is the first year of RBL adoption (see, Coulibaly, 2023; Coulibaly et al., 2022; Mihalyi et al., 2020, 2022), we use it as a deviation date to determine the year of RBL adoption by the non-RBL countries in our sample.

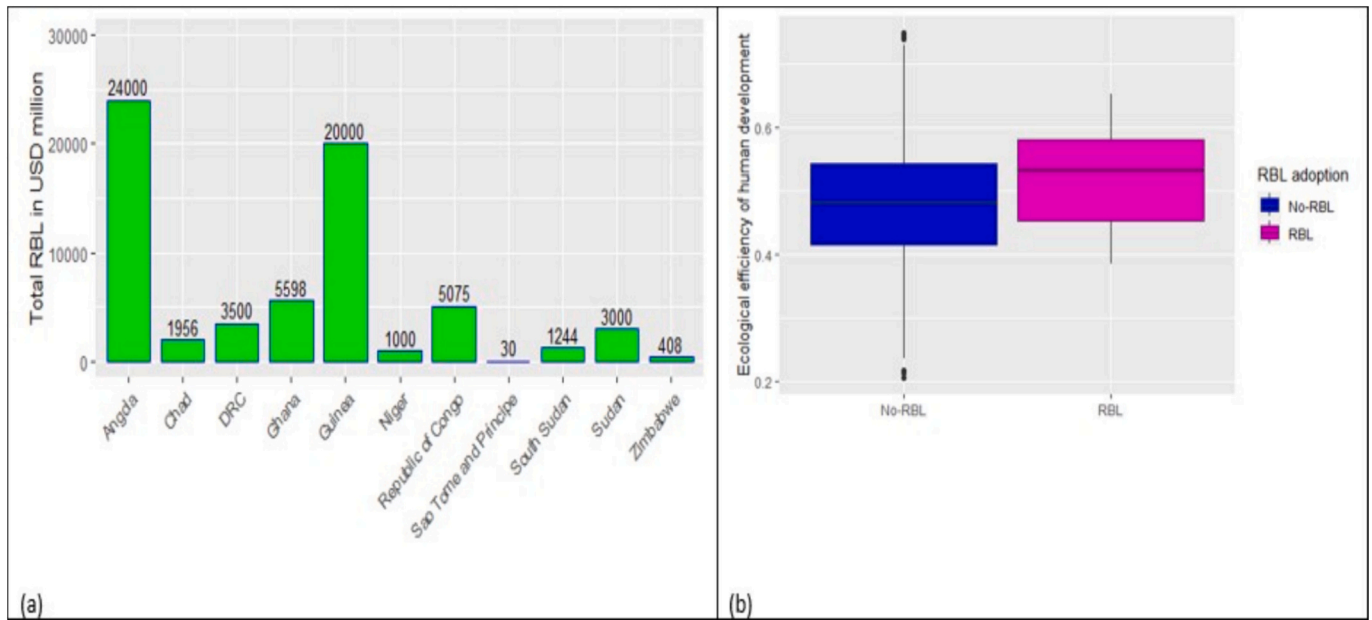


Fig. 1. Total amount of resource-backed loans by country and SDI by RBL adoption.

Source: Author's elaboration using data from NRG, Mihalyi et al. (2020) and Hickel (2020). **Note:** We consider 116 treated observations and 1234 untreated in Fig. 1b. In box plots, the lower and upper hinges of each box show the 25th and 75th percentiles of the samples, the line in the box indicates the respective medians, and the end-points of whiskers mark next adjacent values.

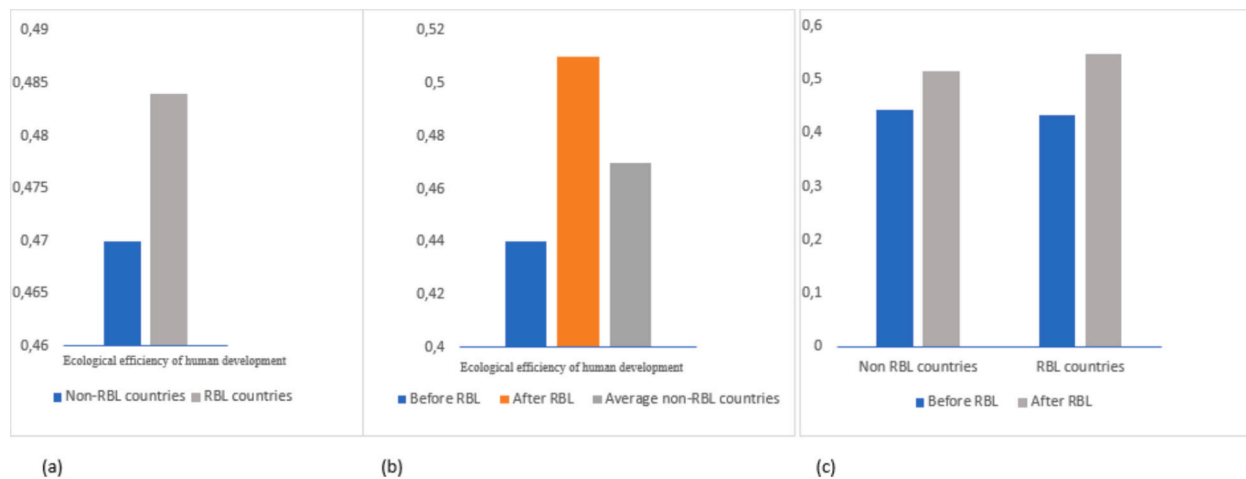


Fig. 2. Sustainable development index and resource-backed loans.

Source: Author's elaboration using data from NRG, Mihalyi et al. (2020) and Hickel (2020).

to take account of long-term deficits and debt, in line with the current state of public finances in most countries of the world. Secondly, they go beyond studies (see, for instance, Barrage (2020); Karydas and Zhang (2019); Fullerton and Kim (2008); Rosendahl (1997); Bovenberg and Smulders (1996), etc.) that limit the exclusive financing of government emission-reduction spending to taxation (i.e. with zero deficit and public debt) by taking public debt into account in their environmental endogenous growth model, and define environmental debt as the gap between natural capital in its pristine state (i.e. the state with no economic activity) and in its current state. To simplify, Boly et al. (2022) argue that in the long term, economic growth is strictly positive and public debt is endogenous, unlike in the OLG context. These authors demonstrate the uniqueness of the steady state and prove that the link between the debt/GDP ratio and the quality of the environment depends on the time horizon. Their conclusions show that public debt and environmental debt are complementary in the long term, and generally

substitute each other in the short term. In other words, an increase in public debt generates a flow of environmentally damaging resources (the public debt burden) and a flow of environmentally improving resources (the public deficit that finances abatement spending). In the long term, the first effect offsets the second, so that public debt has an overall negative effect on pollution abatement spending; whereas in the short term, the second effect outweighs the first, provided that public debt is not too high.

Indubitably, external debt contributes to the depletion of environmental resources on the basis that foreign debt financing and repayment require increased natural resources extraction. So, more debt leads to more resource extraction. The non-reproducible exhaustible environmental resources necessarily exhaust since they cannot regenerate at least in the short run. Tietenberg and Lewis (2018) suggest external debt as a powerful contributor to a rapid depletion of environmental resources. Given that repayment of resource-backed loans can be made

through in-kind resources, this would lead to more excessive mining and affect forest cover. For instance, the exploitation of the Carajás project mine in Brazil has resulted in annual deforestation of 6100 km² as wood from the surrounding forest is converted into charcoal to fuel the smelters (Moran et al., 1994). Similarly, Ranjan (2019) shows that mining contributed to deforestation in 314 districts of India between 2001 and 2014. Kinda and Thiombiano (2021) have also shown that mining and gas rents contribute to accelerating the loss of forest cover. On the other hand, oil rents and resource tax revenues help to reduce the loss of forest cover. In addition, resource-backed loans could undoubtedly increase public spending. This increase in public spending affects deforestation-related activities in developing countries more than non-deforestation-related activities (Combes et al., 2018). However, López et al. (2011), focusing on the composition of public spending, argue that increasing the share of revenues allocated to public and social goods reduces environmental pollution. This argument corroborates with Daniel et al. (2013) that, thanks to effective fiscal policy, revenues from the extractive industries can make a significant contribution to financing productive expenditure. This will strengthen the government's capacity to invest in climate change adaptation and mitigation policies, thereby improving ecological efficiency and environmental quality. Lack of transparency in resource-backed loans can compromise political, institutional, and ecological stability, as corruption increases in many sectors of the economy. Indeed, by attracting extractive industries to the signatory country with massive investments, resource-backed loans offer vast opportunities for corruption and illegal behavior involving significant private capture of resource rents, inducing additional pressure that weakens forest protection. This is the background to the study by Bebbington et al. (2018), who highlighted that corruption accelerates the expansion of extractive industries and is sometimes a direct driver of it. However, mineral-backed resource loans can accelerate the pace of mineral production, which could probably increase the supply of minerals strategic to battery manufacturing for the energy transition. As a result, resource-backed loans could improve environmental quality by accelerating the energy transition. In the spirit of IEA (2019), given the close relationship between climate change and energy consumption, renewable energies (RE) are a global solution for sustainable development. It offers an environmentally friendly, low-emission technology (Malerba, 2019). Based on this argument, it can be argued that natural resource-backed loans offer excellent opportunities for a sustainable development process in line with the UN's development goals and meeting Henderson and Loreau (2023)'s definition. Many infrastructure investments are made with RBL (see Table A9 for more information). It is therefore possible that resources from these RBL investments could be used to finance environmental policies to reduce the forest cover loss (Kinda and Thiombiano, 2021). According to Coulibaly et al. (2022), resource-backed loans reduce the level of public debt after 5-years of adoption and realization of public investments in the projects concerned. This implies that resource-backed loans can provide signatory countries with budgetary leeway to invest in biodiversity and environmental protection, thereby improving ecological efficiency.

At the same time, financing real income growth and energy consumption through external debt can lead to environmental degradation (Katircioglu and Celebi, 2018). Furthermore, countries (especially energy-dependent countries) whose real income growth is influenced by the volume of debt, affect CO₂ emissions through additional energy needs. More explicitly, when foreign debt is invested in real estate, heavy industry and the construction sector, energy demand increases, which can have a negative effect on the environment (Bese, 2021; Sun and Liu, 2020). Drawing a parallel with resource-backed loans, these could therefore increase energy demand and negatively affect the ecological efficiency of development human in adopting countries, since 85% of resource-backed loans amounts are invested in major construction projects such as hydro-electric dams, public buildings, bridges and social housing (Mihalyi et al., 2020).

3.2. Review of the empirical literature

On the empirical side, the few studies that attempt to measure the link between public debt and ecological efficiency or environmental quality concern the evaluation of public debt measured as a percentage of GDP. The consensus is that the link between public debt and environmental quality—most often measured by forest cover—is unambiguously negative. For example, Culas (2006) examines the hypothesis that heavy sovereign debt is the cause of high rates of tropical deforestation, mainly due to the clearing of forest land for agricultural expansion. The results indicate that debt and deforestation are positively linked. Similarly, Didia (2001) and Kahn and McDonald (1995), using classical ordinary least squares regressions, showed that an increase in debt led to an increase in deforestation in tropical countries. In a similar vein, Sommer et al. (2020) looked at the hypothesis of debt reduction and the protection of forest cover. Using the instrumental variables method, the author confirms that larger debt reductions are associated with lower rates of forest loss in the context of debt-for-nature swaps involving the USA. These results are in line with Bese (2021) who estimate the effect of foreign debt on carbon dioxide emissions for China over the period 1974–2014. They find that foreign debt has a positive and significant impact on CO₂ emissions. Indeed, the studies conducted by Bese (2021) and Katircioglu and Celebi (2018) are restricted and country-specific, focusing on emerging countries (Turkey and China, respectively). In contrast to them, Akam et al. (2021) will extend the emerging literature on debt and the environment to a panel of low-income countries to broaden the understanding of the debate. They estimate the impact of external debt on CO₂ emissions for thirty-three (33) heavily indebted poor countries (HIPCs) over the period 1990–2018 using panel data macro-econometric models. The results show that external debt has a positive but not statistically significant impact on CO₂ emissions, suggesting that external debt does not contribute significantly to environmental degradation in HIPCs. Using a propensity score matching—approach on a sample of 64 developing countries over the period 2004–2018, Coulibaly (2023) shows that resource-backed loans contribute to an increase in the level of deforestation in developing countries. The author explains this link through several factors such as natural resource rents, resource depletion, commodity price, and over-indebtedness.

However, it should be noted that policies which do not curb the growth of public debt may reduce the concentration of CO₂ and greenhouse gas emissions and improve environmental quality in the short term. However, it is important to note that such policies can have harmful environmental consequences in the long term. This has been proven by Boly et al. (2022), who use a cointegration technique to establish the link between public debt and environmental debt. The results show that there is a significant and positive long-term link between public debt and environmental debt but may be negative in the short term. These results suggest that public debt could have a positive effect on the environment through green financing. Indeed, public debt can improve the quality of the environment by financing investments in low-carbon technologies, clean energy projects or environmental R&D activities that will primarily benefit future generations (Bénassy-Quéré, 2010). Some authors believe that debt-for-nature swap programs could improve environmental quality and establish fiscal discipline in low-income countries (Nedopil et al., 2023; d Chamon et al., 2022; Essers et al., 2021; Sommer et al., 2020; Cassimon et al., 2011; Thapa, 1998; Kahn and McDonald, 1995; Hansen, 1989; etc.). For instance, Kahn and McDonald (1995) outline a significant positive association between public debt and deforestation in LDCs and defend debt alleviations to reduce the pressure to deforest (Hansen, 1989). According to Gullison and Losos (1993), the public debt could be exchanged for reform of the forestry and agricultural sectors, which would have very significant positive effects on forest conservation and management.

3.3. Testable hypotheses

Since the literature review extensively provides empirical evidence regarding the influence of public debt on environmental quality through CO2 emissions, deforestation. Still, the existing studies are struggling to provide clear evidence as the existing studies come with mixed findings, i.e., some studies have found the positive association, while others found a negative association between the two. Therefore, it is important to reinvestigate this nexus for designing comprehensive economic policy in the African economies. Additionally, a number of studies analyze the association between public debt and human development (see for instance, [Osakede and Adeleke, 2022](#); [Zaghdoudi, 2018](#)). However, no existing studies identify the association between resource-backed loans and human development, which holds significant importance. Even though the studies by [Osakede and Adeleke \(2022\)](#) and [Zaghdoudi \(2018\)](#) retain their quality, they suffer from two major shortcomings. Firstly, their studies do not take into account the typology of public debt, i.e. distinguishing the effect of debts whose repayment directly affects the human development from those that do not. Secondly, their human development index does not consider the ecological sustainability, which is very important for African countries in pursuing and achieving sustainable development goals. This limitation is becoming acutely evident given a growing crisis of climate change and ecological breakdown (see [Hickel, 2020](#)). The use of a Sustainable Development Index (SDI) addressing this concern, i.e. focusing on the ecological efficiency of nations in achieving human development, would have enabled the authors to draw a conclusion and recommendation policies in the direction of the sustainable development process.

The testable hypotheses explored in this paper directly echo the extensive literature reviewed in this section. First, resource-backed loans are expected to have their own importance for the ecological efficiency of human development. *We would therefore reject the null hypotheses that resource-backed loans do not affect the education index, life expectancy index and income index.* Secondly, the literature highlights the direct and indirect effects of public debt on the environmental framework. In particular, RBLs may be less effective in maintaining the boundary planetary at its desired optimal level, unless debtor and creditor governments incorporate an ecological dimension into RBL agreements. *Empirical, this means that we would expect to reject the null hypotheses that the effects of RBL do not influence CO2 emissions and material footprint.* Third and finally, an interesting empirical question arises from the presumption of heterogeneous effects between different types of resource-backed loans. In fact, mining and onshore oil exploitation generate CO2 emissions and destroy forest cover (see [Kinda and Thiombiano, 2021](#)), whereas forests are crucial for carbon sequestration. However, it must be recognized that mineral exploitation could increase the supply of minerals used in battery production for the energy transition and thus reduce global carbon emissions ([Cox et al., 2022](#)). This could mitigate, or even offset, the overall negative environmental impacts of mining compared with those of oil.

Although we have posed null hypotheses in this paper, we do not report the statistical results of these tests for reasons of document size and to avoid confusing the reader. However, tables of results could be provided on request.

4. Methodology

4.1. Motivation and entropy balancing framework

The overall objective of this analysis is to estimate the impact of the adoption of resource-backed loans on ecological efficiency of development in African countries in a more specific context of climate change, energy transition and weak access to capital markets. Our analysis considers that observation is treated when the country has signed a resource-backed loan. Given that most of the countries that have signed up to resource-backed loans are facing numerous macroeconomic and

institutional problems such as sovereign debt default, sovereign debt restructuring, restricted market access and corruption, there are concerns about selection bias, as the adoption of resource-backed loans may be correlated with unobservable variables that may also affect the overall performance of the economy, and thus potentially ecological efficiency of human development. We must use an impact estimation technique that can minimize this selection bias that resides in the RBL adoption policy. To achieve this objective, our study focused on cross-sectional contexts with a binary treatment in which the observational data obtained required covariate adjustments to truly capture the causal effects of RBL on the ecological efficiency of human development. So, a reweighting approach called entropy balancing developed by [Hainmueller \(2012\)](#) was therefore used to make the covariate adjustments and perform a binary treatment. This entropy balancing approach has many advantages and is widely used in economic literature. For instance, [Apeti and Edoh \(2023\)](#) use this approach to assess the impact of mobile money on tax revenue mobilization, [Apeti \(2023\)](#) evaluate the effect of mobile money adoption on household consumption volatility, [Balima and Sy \(2021\)](#) examine the role of IMF-supported programs in reducing the likelihood of subsequent sovereign defaults in borrowing countries, and [Jacolin et al. \(2021\)](#) investigate the impact of mobile financial services in particular, mobile money, mobile credit, and savings on the informal sector. [Kinda and Thiombiano \(2024\)](#) examine the implementing the Extractive Industries Transparency Initiative (EITI) standard on deforestation in resource-rich developing countries. The approach used in this study is based on the principle that the adoption of resource-backed loans is the treatment, and that ecological efficiency of human development is the outcome variable. The units of observation are country-year. Observations with RBL constitute the treatment group, and those without RBL constitute the control group. The treatment effect on treated individuals (ATT) is defined as follows:

$$ATT = E[Y_{(1)} | T = 1] - E[Y_{(0)} | T = 1] \quad (1)$$

Where $Y_{(j)}$ is the outcome variable measuring the ecological efficiency of human development. T indicates whether the observation unit is subject to resource-backed loans adoption ($T = 1$) or not ($T = 0$). $E[Y_{(1)} | T = 1]$ is the ecological efficiency of human development level during the resource-backed loans adoption period, $E[Y_{(0)} | T = 1]$ is the counterfactual result for countries that had adopted resource-backed loans, i.e. the ecological efficiency of human development in countries that had adopted resource-backed loans if they had not.

The issue is that $E[Y_{(0)} | T = 1]$ is not observable due to the non-random nature of RBL adoption. If this were the case, the ATT could easily be identified by comparing the ecological efficiency of human development in RBLs countries with non-RBLs countries. Identifying ATT then requires a good proxy for $E[Y_{(0)} | T = 1]$. To do so, we match RBL units with non-RBLs units (after purging for some specific factors) that are as close as possible on observable characteristics that meet two criteria: correlated with RBL adoption and ecological efficiency of human development. Under the condition that the non-RBLs units are fairly close to the RBLs units, any difference in consumption volatility is attributable to RBL adoption. Based on these details, we can rewrite Eq. (1) as follows:

$$ATT = E[Y_{(1)} | T = 1, X = x] - E[Y_{(0)} | T = 0, X = x] \quad (2)$$

where $X = x$ is a vector of observable covariates that may affect both the decision to adopt RBL and the ecological efficiency of human development, $E[Y_{(1)} | T = 1, X = x]$ is the ecological efficiency of human development of RBL units, and $E[Y_{(0)} | T = 0, X = x]$ is the expected ecological efficiency of human development for the synthetic control units. Entropy balancing is a two-step estimation method. First, the weights assigned to the control units (in this case, the non-RBL countries) are calculated in the entropy balancing. In the second step, the weights obtained in the first step are used in a regression analysis with the treatment variable (RBL countries) as the explanatory variable. We

then weigh the countries with RBL and non-RBL based on observable characteristics. Thus, the average difference in ecological efficiency of human development between countries with RBL and the “closest” non-RBL countries should be explained by the RBL adoption.

4.2. Advantages of entropy balancing

Unlike other methods for estimating effects and treatment effects, such as propensity score matching even difference-in-difference, entropy balancing has several advantages because of its ability to combine both matching and regression analysis (Apeti, 2023; Balima and Sy, 2021; Sawadogo, 2020; Neuenkirch and Neumeier, 2016; Hainmueller, 2012). In fact, this method outperforms the classical approach based on regression and matching based on propensity score methods because it is non-parametric—which allows to avoid the problem of mis-specification of the functional form of the model that could bias the results. Moreover, it is a method that also eliminates multicollinearity problems since the mechanism of re-weighting makes the treatment variable orthogonal to the covariates (Hainmueller, 2012). Another advantage is that entropy balancing uses more flexible reweighting schemes compared to conventional balancing, where control units are either discarded or balanced. Here, units are reweighted with the goal of achieving a balance between processed and unprocessed units, keeping the weights as close as possible to the base weights to avoid information loss. Its most attractive feature is to allow a high degree of covariate balance between RBL countries and non-RBL countries—even in small samples—by creating a synthetic control group that is as close as possible to the program group. Finally, classical matching methods and pooled probit models rely on the assumption of conditional independence, i.e., based on a vector of observable covariates, the treatment is independent of unobservable factors.

In addition, the use of entropy balancing allows us to account for the panel dimension of the data by controlling for country—and time-specific factors in the second stage of the regression analysis. It is necessary to include country-specific effects to account for potential unobserved heterogeneity between countries that have never signed RBL and those that have. Indeed, the macroeconomic environment of these two groups may differ beyond the covariates used in the entropy balancing approach. Country effects also control for time-invariant, country-specific conditions that may lead to differences in ecological efficiency of human development between countries. However, entropy balancing may not control for potential endogeneity biases due to unobserved temporal factors that may affect both RBL and ecological efficiency of human development, as well as the reverse causality problem that may exist between the treatment variable and the outcome variable and cannot successfully deal with the inertia of ecological sustainability and the pursuit of sustainable development goals. This can lead to potentially biased results. For this reason, we complement entropy balancing with alternative estimation methods (see, section 7).

5. Data

5.1. Treatment variable

The treatment variable used in this study is RBL adoption, a binary that variable we constructed based on information contained in the NRG (Natural Resources Governance Institute) database. So far, NRG remains the only institution that provides reliable and solid evidence on the existence and many details of resource-backed loans in developing countries. A team of researchers led by Mihalyi, D., Adam, A. and Hwang, J. collected as much information as possible on resource-backed loans to obtain a somewhat larger database. They compiled two original datasets maintained by institutions dedicated to researching China's overseas activities: the China-Africa loans dataset produced by the China-Africa Research Initiative (CARI) at Johns Hopkins SAIS and the China-Latin America financial database of the Inter American Dialogue

and Global Development Policy Center at Boston University (Mihalyi et al., 2020). Next, the authors supplemented these databases with additional literature searches to present their dataset. It should be noted that the data on this resource-backed loans are not fully comprehensive due to the relative opacity of these transactions (Rivetti, 2021). There could certainly be other resource-backed loans in both regions than those listed by the authors as they are limited by publicly available information (Mihalyi et al., 2020). However, it is the most comprehensive dataset on resource-backed loans for African and Latin America economies over a long period. Following Coulibaly (2023); Coulibaly et al. (2022); Mihalyi et al. (2020); Mihalyi et al., 2022); Bräutigam and Gallagher, 2014), we construct a dummy variable that takes the value 1 if country i at a period t has taken out a resource-backed loan and 0 otherwise.

5.2. Outcome variable

The dependent variable used in this paper is captured by the new sustainable development index constructed by Hickel (2020). To construct this index, Hickel (2020) starts from the premise that the Human Development Index (HDI) introduced in the 1990s and modified by UNDP⁴ is not a perfect measure for millennium goals, and that it was necessary to implement a new method, technique and methodology to build an index that would take into account the limitations of the HDI that have become evident in the 21st century, given the growing crisis of climate change and ecological degradation.

In fact, before Hickel (2020), numerous studies have attempted to propose indicators for correcting the HDI (O'Neill et al., 2018; Raworth, 2012, 2017; Steffen et al., 2015; Gough, 2015; Rockström et al., 2009), particularly those that cross ecological boundaries (Morse, 2003; Neumayer, 2001; de La Vega and Urrutia, 2001; Neumayer, 2001; Sagar and Najam, 1998; Ramanathan, 1999), other studies have looked at how the environmental dimension is taken into account in the design of the HDI (Hirai, 2017), more specifically, to integrate CO2 emissions per capita (Bravo, 2014), incorporates material footprint (Zhang and Kai, 1998). However, not all these studies address the question of human development in its entirety, and all suffer from notable weaknesses that make the m ultimately unsuitable (Hickel, 2020). As a result, Hickel (2020) proposes an index with two specific objectives namely: correct the HDI according to its ecological impact, and propose a simple alternative that retains the logic and basic structure of the original.

Basically, the sustainable development index proposed by Hickel (2020) comprises five indicators: education, life expectancy, income, CO2 emissions and material footprint. The formula is as follows:

$$SDI = \frac{\text{Development Index (DI)}}{\text{Ecological Impcat Index (EII)}} \quad (3)$$

The development index is calculated as the geometric mean of the education index, the life expectancy index, and the modified income index. This gives the following formula:

$$DI = \sqrt[3]{\text{Educatin Index} * \text{Life Expectancy Index} * \text{Income Index}} \quad (4)$$

In this formula, the education index and the life expectancy index are like the HDI developed by Mahbub ul Haq (UNDP, 1997). So, we obtain the income index using the following formula:

$$\text{Income Index} = \frac{\ln(\text{GNIpc}) - \ln(100)}{\ln(20,000) - \ln(100)} \quad (5)$$

Concerning the denominator of Eq. (3), ecological impact index, it is obtained using the following formula:

$$EII = 1 + \frac{e^{AO} - e^1}{e^4 - e^1} \quad (6)$$

⁴ See, Kovacevic et al. (2010)

We can see that if $AO > 4$, then $EII = AO - 2$. This is how we obtain the AO formula:

$$AO = \sqrt[2]{\left(\frac{MF}{boundary} \geq 1\right) * \left(\frac{CO2}{boundary} \geq 1\right)} \quad (7)$$

The author proceeds to a kind of AO normalization by dividing the material footprint and emissions values by per capita planetary boundary. The idea behind this task is that a country cannot compensate for overshooting one boundary by undershooting the other, because overshooting either boundary will result in an average overshoot >1 . After the calculation, [Hickel \(2020\)](#) builds a database with 163 countries having data points for the five indicators included in the IDS from 1990 to 2019.

However, the sustainable development index devised by [Hickel \(2020\)](#), while incorporating new perspectives in the improvement of the HDI, may suffer from the bias of missing relevant variables. In fact, the author does not include in his calculation certain indicators linked to basic physical needs, such as access to food, electricity, improved sanitation and housing, literacy, and child mortality, all of which are targeted by the SDGs. Despite this, we have to acknowledge that it is the only indicator that captures both development, economic growth and environmental issues, and has data for many countries over a long period, unlike the other indicators, for instance, the one designed by [O'Neill et al. \(2018\)](#). Data are directly collected on Sustainable Development Index.

5.3. Controls or matching variables

Following the literature, we control several variables that are likely to have an impact on environmental sustainability. These variables are inflation rate, ODA, foreign direct investment, natural resource rents, and resource depletion. For robustness, other relevant variables will be controlled in future sections.

We measure the **inflation rate** using the consumer price index. The use of this variable is justified by the fact that there is a direct relationship between the inflation rate and the environment ([Ronaghi et al., 2020](#)). [Ronaghi et al. \(2020\)](#) has shown that inflation has a statistically significant negative impact on CO2 emissions. It is well known that higher inflation is accompanied by higher commodity prices in the country. As a result, consumer demand for goods decreases, and by reducing production, CO2 emissions eventually decrease. This is a boon for environmental sustainability. Another reason could be that inflation reduces trade, so CO2 emissions will fall indirectly.

However, both effects are observed with **foreign direct investment (FDI)**. According to [Pazienza et al. \(2015\)](#), there is a negative relationship between foreign direct investment (FDI) and CO2 emissions. Basically, higher levels of foreign direct investment and employment will increase production, leading to more emissions ([Ronaghi et al., 2020](#)). Likewise, [Lee \(2013\)](#) believes that FDI flows are a catalyst for economic growth through technology transfer, the introduction of new processes and productivity gains, as well as management skills. Under these conditions, technology transfer, for example, could generate economies of scale, accelerate the energy transition, and improve environmental sustainability. Taking a closer look at oil-exporting countries, FDI is a major factor in environmental degradation. This is because higher levels of economic activity (production and consumption) require more energy and raw materials, thus increasing CO2 emissions.

Official development assistance (ODA) could have a rather positive effect on the environment since donors and development partners actively support developing countries in their quest for environmentally friendly development. This corroborates the results found by [Kwakwa et al. \(2020\)](#) which showed that official development assistance contributes to reducing long-term carbon dioxide emissions in Ghana.

Natural resources rents are the sum of oil rents, natural gas rents, coal rents, mineral rents, and forest rents. The effect of natural resource

extraction on the environment is somewhat ambiguous in the literature. While the results of the study reveal that natural resource rents improve the natural environment by reducing carbon emissions ([Joshua and Bekun, 2020](#); [Wang et al., 2020](#)), other results reveal that natural resource rents cause more carbon emissions ([Anh Tu et al., 2023](#); [Kinda and Thiombiano, 2021](#)).

Natural resource depletion⁵ is the sum of net forest depletion, energy depletion, and mineral depletion. Resource depletion can have a devastating impact on the environment quality. For example, mining can lead to habitat destruction, pollution of water bodies and soil erosion. These environmental impacts can have considerable effects on ecosystems and biodiversity. Due to soil erosion and soil fertility are considerably affected and rendered useless, resulting in the drying out and aridity of certain parcels of land over time, and thus leading to a loss of wildlife and vegetation. On the other hand, resource depletion can call on political decision-makers to implement adaptation strategies, sustainable investments, and biodiversity protection for future generations, given that depleted resources are generally non-renewable. This can undermine the sustainability of development in these countries.

6. Results

6.1. Descriptive statistics

We present the descriptive statistics for the first-stage equation in [Table 1](#). This is a simple comparison of sample means before weighing for all matching covariates between treated units (column [1]) and control units (column [2]), which represent the potential synthetic group.

[Table 1](#) shows that, on average, the treated observations report lower inflation and receipt of official development assistance, greater attraction of FDI and natural resource rents, and a higher rate of resource depletion. Next, we note that the differences between the two groups are significant (column [5]), except for the inflation rate, which could bias the effect of the measure due to a potential selection problem if we estimate the treatment effect using traditional econometric methods such as OLS estimators. To overcome this problem, we construct a synthetic

Table 1
Descriptive statistics before weighting.

	[1] RBL	[2] Non RBL	[3] = [2]– [1] Difference	[4] t- Test	[5] P- value
Log Inflation rate	1.91	1.80	–0.11	0.5	0.489
Log ODA	1.41	1.74	0.33	4.2	0.040
Log FDI	1.57	0.53	–1.04	30.3	0.000
Log Natural resources rents	2.56	1.75	–0.81	17.9	0.000
Log Resource depletion	2.05	0.87	–1.18	18.6	0.000
Observations	240	1110			

This table shows the sample means before weighting for matching covariates for country-year observations where RBLs were signed (the treatment group) in column [1] and country-year observations where no RBLs were ever signed (the potential control group) in column [2]. Column [3] shows the differences in means between the treatment and control groups, while the corresponding *t*-test statistics and *p*-values are columns [4] and [5] respectively.

⁵ Net forest depletion is unit resource rents times the excess of round wood harvest over natural growth. Energy depletion is the ratio of the value of the stock of energy resources to the remaining reserve lifetime (capped at 25 years). It covers coal, crude oil, and natural gas. Mineral depletion is the ratio of the value of the stock of mineral resources to the remaining reserve lifetime (capped at 25 years). It covers tin, gold, lead, zinc, iron, copper, nickel, silver, bauxite, and phosphate.

control group by reweighting the control units using the pre-treatment covariates of the baseline specification. In fact, this approach makes the pre-treatment covariates of the synthetic group on average as comparable as possible to those of the treated units based on several similar characteristics. Table 2 shows in columns [1] and [2], the sample mean after weighting between the treatment group and the synthetic group obtained by entropy balancing, and column [3] shows the difference between the first two. Analysis of the two groups in this table reveals the effectiveness of entropy balancing, as the difference shown in the previous table seems to disappear. Consequently, we can consider the synthetic group as an «near perfect» counterfactual of the treated group (RBL countries).

6.2. Treatment effects

Table 3 presents the average treatment effect on RBL countries for the ecological efficiency of human development, obtained from different sets of treatment effect estimates. First, in columns 1 to 4, we present the basic results highlighting the effect of resource-backed loans adoption on sustainable development. Next, we add a country fixed effect, a time fixed effect and country/year fixed effects. In columns [5]–[8], we include all control variables in our baseline regression. The results indicate that the adoption of a resource-backed loan significantly increases the ecological efficiency of human development. The magnitude of the coefficients suggests that adopting resource-backed loans improves the ecological efficiency of human development by around 2.41 percentage points. These results are robust; whatever the specifications, the estimated effects are positive and statistically significant. In addition, when the country signs a resource-backed loan, the ecological efficiency of human development varies by up to 1.2% more than countries observed without a resource-backed loan (these two groups of countries are similar in terms of pre-treatment factors). This effect is statistically significant at 1%, especially when we include control factors. These results could be explained by the fact that countries taking out resource-backed loans have taken advantage of RBL to invest in their energy sector, road infrastructure, healthcare, and education. In addition, several social housing construction projects have been carried out thanks to resource-backed loans. According to Mihalyi et al. (2020), governments allocated over 85% of RBL and over 90% of the value of RBL to capital expenditure. Given that achieving development goals contributes to improving the well-being of the population, raising the education index, life expectancy and income levels will increase the development index. As a result, resource-backed loans will eventually have a positive impact on the ecological efficiency of human development. The positive impact of resource-backed loans on the ecological efficiency of human development suggests that policy-makers in borrowing countries might consider negotiating with creditors of resource-backed loans with a focus on public investment. However, they will need to include in loan agreements key provisions for the financing of biodiversity components and the implementation of actions to

mitigate the negative environmental impact of resource-backed loans, which should mitigate the ecological impact that is in the denominator of ecological efficiency of human development calculate by [Hickel \(2020\)](#).

Our results are in perfect harmony with and even corroborate those already found by some, who believe that nations with abundant subsoil assets in the form of oil, gas and minerals can derive significant revenues from extraction that can be mobilized for economic development ([Viner, 1952](#); [Hsu et al., 2023](#); [Carbonnier et al., 2011](#); [Brunnschweiler, 2008](#); [Hordofa et al., 2022](#)). Intentionally, we have decided not to report covariate coefficients in the tables, but a complete table with covariate signs and coefficients can be provided on request. Then, in [section 7](#), we decided to confound our results with a series of tests to ensure that these results are not affected by any modification.

7. Robustness checks

Our estimates show that the adoption of resource-backed loans significantly increases the ecological efficiency of human development. In this section, we test the robustness of these results with the aim of strengthening our confidence.

7.1. Alternative estimation method

7.1.1. Difference-in-differences estimators

The difference-in-differences (DiD) method is a more robust and most popular method used to assess the causal effects of policy interventions using observational data ([Callaway and Sant'Anna, 2021](#); [Sant'Anna and Zhao, 2020](#)). In its canonical or basic form, it is based on two periods and two groups. In the first period, no units are treated, and in the second period, some units are treated (treatment group) and some units are not (comparison group). In this case, if the average scores of the treated and comparison subjects without treatment would have evolved in parallel over time (parallel trends hypothesis), we can estimate the average treatment effect for the treated sub population (ATT) by comparing the average change in scores observed by the treated group with the average change in scores recorded by the comparison group ([Callaway and Sant'Anna, 2021](#); [Sant'Anna and Zhao, 2020](#); [Botosaru and Gutierrez, 2018](#); [Abadie, 2005](#); [de La Vega and Urrutia, 2001](#)). There are several reasons for using this technique. Firstly, we are exposed to a phased adoption problem ([Sant'Anna and Zhao, 2020](#)), i.e. in our dataset, once countries have signed a resource-backed loan agreement, they continue to benefit from resource-backed loans in subsequent periods, as the maturity of resource-backed loans for borrowing countries goes beyond the last observation period of the panel we are working on (the year 2019). In this case, we are dealing with average treatment effects with several time periods and parallel trends that are potentially valid only after conditioning on the observed covariates. Indeed, we have used entropy balancing for the basic regression, which uses linear regression (OLS) in its second stage. This estimate, which includes unit-specific time averages and period-specific cross-sectional averages, is referred to as the Two-Way Fixed Effects (TWFE) estimator by [Wooldridge, \(2021\)](#).

However, the empirical literature on impact evaluation methodologies shows that the TWFE estimator produces biased estimates when the treatment effect is heterogeneous within countries due to negative weights in the calculation of the average treatment effect ([Athey and Imbens, 2022](#); [Sun and Abraham, 2021](#); [Nichols et al., 2019](#); [Borusyak and Jaravel, 2018](#); [de La Vega and Urrutia, 2001](#)). This seems to be the case in our context, where some countries are accumulating resource-backed loans. In addition, the second reason for using difference-in-difference is that there may be endogeneity in the resource-backed loans variable, as countries with poor ecological efficiency of human development performance may face a higher risk premium for borrowing, and therefore may resort to resource-backed loans. Finally, to circumvent the potential endogeneity problem of resource-backed

Table 2
Descriptive statistics after weighting.

	[1]	[2]	[3] = [2]–[1]
	RBL	Non RBL	Difference
Log Inflation rate	1.906	1.906	0.000
Log ODA	1.407	1.407	0.000
Log FDI	1.57	1.569	–0.001
Log Natural resources rents	2.563	2.562	–0.001
Log Resource depletion	2.049	2.048	–0.001

This table shows the sample means after weighting for matching covariates for country-year observations where RBLs were signed (the treatment group) in column [1] and country-year observations where no RBLs were ever signed (the potential control group) in column [2]. Column [3] shows the differences in means between the treatment and control groups.

Table 3

Resource-backed loans and ecological efficiency—Baseline results.

VARIABLES	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
Resource-backed loans	0.109*** (0.0205)	0.163*** (0.0132)	0.0515*** (0.0194)	0.0286*** (0.00797)	0.0831*** (0.0176)	0.114*** (0.0123)	0.0315* (0.0165)	0.0241*** (0.00718)
Constant	−0.695*** (0.0101)	−0.768*** (0.0284)	−0.743*** (0.0668)	−0.775*** (0.0390)	−0.503*** (0.0254)	−0.629*** (0.0409)	−0.525*** (0.0642)	−0.696*** (0.0455)
Covariates	No	No	No	No	Yes	Yes	Yes	Yes
Country/FE	No	Yes	No	Yes	No	Yes	No	Yes
Year/FE	No	No	Yes	Yes	No	No	Yes	Yes
Observations	799	799	799	799	799	799	799	799
R-squared	0.098	0.813	0.214	0.938	0.394	0.854	0.486	0.940

This table presents the effect of resource-backed loans on sustainable development obtained by weighted least squares regressions. The treatment variable is the presence of a resource-backed loan. The outcome variable is the occurrence of ecological efficiency of human development. The control variables include inflation rate, ODA, FDI, natural resource rents, and resource depletion. Robust standard errors are in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

loans, we follow Callaway and Sant'Anna (2021) who develop a difference-in-difference (DiD) estimator not only for staggered adoption but also to address the potential endogeneity problem. Indeed, the advantage of this difference-in-differences estimator is that it effectively bypasses the negative weight problem, the endogeneity problem associated with the TWFE estimator and estimator and considers the heterogeneity and dynamics of treatment effects (Callaway and Sant'Anna, 2021).

To achieve this, we estimate three different types of DiD estimators under staggered treatment adoption always following Callaway and Sant'Anna (2021). The first is based on outcome regressions (Heckman et al., 1997; Heckman et al., 1998a), the second is based on doubly robust methods (Sant'Anna and Zhao, 2020), and the third based on inverse probability weighting (Abadie, 2005). Finally, in the regressions, we consider observations never processed and those not yet processed as a control group (see, for instance, Athey and Imbens, 2006, De Chaise-martin and d'Haultfoeuille, 2018).

We report the results in Table 4. In column [1], we present the doubly robust DiD estimator based on stabilized inverse probabilistic weighting and ordinary least squares (dripw) without covariates. In column [3], we report the results of the inverse probability weighted DiD estimator (ipw), also without covariates. In columns [3]–[4], we repeat the same regressions, augmenting our model with the covariates used in the base model. The overall treatment effect (ATT) is positive and statistically significant, suggesting that the main results of the study remain qualitatively unchanged when estimates are performed with DiD estimators.

7.1.2. Propensity scores matching estimates (PSM)

The principle of the PSM method is to match countries that have signed RBLs with countries that have not signed but share certain observable characteristics linked to both signing an RBL and ecological efficiency. These observable characteristics are summarized in a propensity score (PS) which indicates the probability of a country signing an RBL (treatment group), subject to the observable characteristics (Smith and Todd, 2005; Dehejia and Wahba, 2002; Heckman et al., 1997; Rosenbaum and Rubin, 1983). The propensity score is then used to determine a group of countries that have not signed an RBL (control group). This group of countries serves as a counterfactual for the treatment group. Indeed, assuming that the determinants of ecological

efficiency of human development are statistically independent of the RBL signature, given the common characteristics between the treatment and control groups, the difference in outcome between the two groups, the average effect of treatment on treated (ATT), is attributable to the RBL adoption.

In addition, this method consists of two steps. First, we calculate a propensity score (PS) using a logit model with RBL as the dependent variable. Following Apeti et al. (2024); Combes et al. (2024); Apeti et al. (2023); Chapel (2022); Smith and Todd (2005); Dehejia and Wahba (2002); Heckman et al. (1997); Rosenbaum and Rubin (1983), we use different matching algorithms to test the robustness of our results. The matching techniques used include nearest-neighbor matching with replacement, which matches each treated country to the n control countries with the closest PS (we consider $n = 1$, $n = 2$ and $n = 3$). Next, radius matching is also used to match an RBL country to non-RBL countries whose PS lies within a radius of length r (we consider a wide radius $r = 0.045$, a medium radius $r = 0.09$ and a narrow radius $r = 0.18$). Next, we follow Fan (1993) and Heckman et al. (1998b) in using regression-adjusted local linear matching to couple covariate-adjusted results for the treatment group with corresponding covariate-adjusted results for the control group using local linear regression weights. Finally, we use kernel matching as a matching algorithm to associate a treated country with all control countries, weighted proportionally to their proximity in terms of PS to the treated country. Before interpreting the results, we perform several validation tests. First, we follow Dehejia and Wahba (2002) and calculate standard errors using the bootstrap technique to obtain robust standard errors for valid inference. Next, we use the diagnostic tools of Rosenbaum and Rubin (1985) to check the validity of the conditional independence hypothesis (no evidence of significant differences on observable characteristics within the matched data between RBL and non-RBL countries), and hence the validity of the matching results. The conditional independence assumption holds when the p -value associated with its standardized bias score t -test statistics is $> 5\%$. In addition, we use Rosenbaum (2002)'s sensitivity tests to check the extent to which unobserved heterogeneity could alter the results of the RBL impact estimate on ecological efficiency of human development. We report the new ATTs in Table 5, along with the tests for matching quality, while the results of propensity score estimation are presented in the Table 6, column [1]. Firstly, the results are stable and confirm the positive effect of RBLs on ecological efficacy of human development. Secondly, all the Pseudo-R² of our estimates are below 10%, suggesting that the matching provided balanced scores. In other words, our results are robust to the common support hypothesis.

7.1.3. OLS estimates

After evaluating the robustness of our results with the impact estimates (PSM), we re-estimate our model using the Ordinary Least Squares (OLS). The results presented in column [1]–[6] of Table A1 in Appendix suggest a positive and significant effect of RBL on ecological efficiency of

Table 4

Resource-backed loans and ecological efficiency: Diff-in-Diff estimations.

	[1]	[2]	[3]	[4]
	dripw	ipw	dripw	ipw
ATT	0.0437** (0.0170)	0.0438** (0.0170)	0.0071* (0.0039)	0.0392*** (0.0115)
Covariates	No	No	Yes	Yes
Observations	1044	1044	654	654

Standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5

Resource-backed loans and ecological efficiency: using PSM.

Dependent Variable:	1-Nearest Neighbor	2-Nearest Neighbor	3-Nearest Neighbor	Radius Matching		Local linear regression		Kernel
Ecological efficiency		Matching		r = 0.045	r = 0.09	r = 0.18	Matching	Matching
ATT	0.0855** (0.0404)	0.0916*** (0.0312)	0.0904*** (0.0350)	0.0931*** (0.0221)	0.0998*** (0.0287)	0.1069*** (0.0251)	0.0835*** (0.0299)	0.0949*** (0.0286)
Observations	799	799	799	799	799	799	799	799
Treated	65	65	65	65	65	65	65	65
Untreated	734	734	734	734	734	734	734	734
Matching Quality	0.000	0.007	0.005	0.006	0.024	0.078	0.000	0.006
Pseudo-R2								
Rosenbaum bounds sensitivity test	1.55	2.9	1.85	3.00	3.00	3.00	3.00	3.00
Standardized bias (p-value)	1.000	0.944	0.967	0.964	0.529	0.015	1.000	0.964

Standard errors in parentheses *** p < 0.01, ** p < 0.05, * p < 0.1.

Table 6

Results from Propensity Score, GMM and IV-2SLS estimators.

VARIABLES	[1] Propensity Score	[2] GMM	[3] IV-2SLS
Lag Ecological efficiency		0.4176*** (0.0332)	
RBL		0.0148** (0.0070)	0.7963*** (0.2563)
Log Inflation	0.0300 (0.1030)	−0.0025*** (0.0008)	0.0050 (0.0074)
Log ODA	−0.2256*** (0.0727)	−0.0357*** (0.0024)	−0.0549*** (0.0084)
Log FDI	0.4983*** (0.1307)	0.0091*** (0.0009)	0.0044 (0.0066)
Log Natural resources rents	0.3504*** (0.1298)	−0.0193*** (0.0014)	−0.0454*** (0.0062)
Log Resource depletion	0.0789 (0.0787)	0.0152*** (0.0010)	0.0189*** (0.0041)
Constant	−3.5461*** (0.2808)	−0.3336*** (0.0191)	−0.6186*** (0.0195)
AR(1) test, p-value		0.005	
AR(2) test, p-value		0.969	
Hansen, p-value		0.682	
Sargan, p-value		0.000	
Kleibergen-Paap LM test, p-value			0.0043
Observations	926	776	716
R-squared	0.1190		
Centered R2			−0.2156
Uncentered R2			0.9068
Number of countries		38	45
Number of instruments		20	2

Robust standard errors in brackets. * p < 0.1, ** p < 0.05, *** p < 0.01.

human development, with a magnitude of around 3.22 percentage points, qualitatively comparable to that of the main model (2.4 percentage points, Table 3, column [8]).

7.1.4. System-GMM estimates

We discuss the econometric problems that can arise when estimating the baseline specification. In fact, the ecological efficiency of human development index is likely to be endogenous, due to the reverse causality of RBL to ecological efficiency of human development. Indeed, countries with a low ecological of human development index could be tempted to sign an RBL to finance sustainable development and combat poverty and income inequality. In addition, endogeneity may also arise due to measurement error in the ecological efficiency of human development index, since the ecological efficiency data we used in this study was estimated by [Hickel \(2020\)](#). We follow the literature on dynamic panel data estimation to solve these endogeneity problems using the system-GMM estimator ([Blundell and Bond, 1998](#)). This method has

several advantages. It controls for the persistence of ecological outcomes, notably ecological efficiency of human development, it controls for the Nickell bias ([Nickell, 1981](#)) that appears in a dynamic panel model - a model with lagged endogenous control variables - with fixed effects, it limits the influence of unobservable time-varying factors that can influence both outcome and treatment variables, and it alleviates the difficulty of finding an exogenous instrument to estimate the effect of RBLs. One of the main advantages of the systemic GMM estimator lies in the fact that this estimation technique can instrument other explanatory variables that could potentially be endogenous in addition to the main regressor (in our case, RBL), which is endogenous. However, one limit of the GMM estimator is that lagged levels of variables may be poor instruments (weak instruments problem) if those variables follow a random walk process as lagged values of the variables in level provide less information about future changes ([Blundell and Bond, 1998](#)). Moreover, this method is considered by some researchers to be a black box (see [Soto, 2009](#)), in the sense that it offers researchers the possibility and ease of manipulating the number of lag until they obtain the desired results.

We present the results in column [2] of Table 6. Unsurprisingly, the results lead to conclusions that are qualitatively like the baseline results. In addition, regarding the instrument selection criteria, the Hansen test does not reject the hypothesis of instrument validity. Finally, the AR (1) test for the absence of autocorrelation of the first-order error term and the AR (2) test for the absence of autocorrelation of the second-order error term do not raise concerns about the validity of our estimates.

7.1.5. Instrumental variables estimation (IV)

In addition to the aforementioned robustness checks, it would also be reassuring to display the system's GMM estimation results obtained with external instruments in addition to the internal ones, in order to mitigate the RBL endogeneity problem. Commodity prices and crude oil prices are liable to affect the probability of signing a resource-backed loan, without having any direct effect on the ecological efficiency of human development of African economies. Consequently, commodity prices and crude oil prices could be good candidates (not perfect) as instrumental variables for RBL. Although method IV uses external instruments not only to overcome the problems of instrument proliferation and the black-box problem highlighted in the literature, but it also has its limitations. The main limitation of this method is that in theory it is difficult, if not impossible, to obtain a perfect external instrument, i.e., one that is correlated with the endogenous variable without impacting the dependent variable.

We present the results in Table 6, column [3]. The main result of the study remains qualitatively unchanged when systemic GMM estimation includes an external instrument. Resource-backed loans have a positive impact on ecological efficiency.

7.2. Additional control variable and modification of sample

We examine the robustness of the main result of the paper when various considerations are taken onboard in the analysis. We have grouped these variables into four categories defined as follows: *i*- economic variables: GDP per capita volatility, investment volatility, total investment, stock price volatility, financial openness, rich-resources, trade openness, remittance inflows, public debt, exchange rate; *ii*- political and institutional conditions: rule of law, corruption, external conflict, internal conflict, political stability; *iii*- climate considerations: climatic vulnerability, disasters index, exposure index, shock index, natural shock index; *iv*- infrastructure and structural issues: ICT infrastructure, education, innovation, doing business, electric power consumption, access to electricity urban, access to electricity rural. To capture the common characteristics of the variable categories used, we calculate a composite outcome index for each category following Anderson (2008), in order to apply generalized least squares estimators. The steps and advantages of this method will be discussed in the appendix.

Indeed, we begin this exercise by including these four groups of variables one at a time in our baseline model, thus mitigating any potential multicollinearity problems we may encounter. The results in Table A2, columns ([1]–[5]) in Appendix present the conclusions from the inclusion of economic variables, political and institutional conditions, climatic considerations, and infrastructural and structural factors respectively. The principals remain unchanged when these four categories of variables are included in the baseline equation, despite some loss in the magnitude of coefficients and observations compared to the Table A1 in Appendix. However, despite the consistency of our results, they may potentially suffer from omitted-variable bias, as we conveniently decided to include these four variables separately in our regressions. Consequently, in column [5], we simultaneously include our four previous groups in the base model. The findings remain consistent with those reported previously. Next, we disaggregated each variable category to run the regressions, allowing us to capture the effect or modification that each variable used to construct the variable categories exerts on our baseline results. We report the new results in Tables [A3]–[A4], columns [1]–[27] in the Appendix. The conclusions remain unchanged and robust, resource-backed loans have a positive and statistically significant impact on ecological efficiency of human development.

Finally, we take the rather crude step of modifying our overall sample and testing the robustness of our results. To achieve this, we first run regressions on different sub-samples. Following Lin and Ye (2013) and Reinhart and Rogoff (2004), we exclude hyperinflation episodes defined as an annual inflation rate of 40% or more. Secondly, we exclude newly signed RBLs from our sample. In fact, new RBLs may not have an effect on the ecological efficiency of human development because of the time it takes for an investment to have a significant effect on the economy. In this way, RBLs may not have effect in Guinea, which signed up in 2017 and Ghana in 2018, for example, as our sample is limited to 2019. We limit our sample to 2004. This allows us to exclude all observations from before 2004, which is considered to be the starting point for the adoption of RBL in African countries (Mihalyi et al., 2020, 2022). In addition, we exclude the collapse of the world economy due to the 2008–2009 financial crisis, as well as the year of the oil crises. Some countries are actively participating in various environmental conventions and adopting environmentally friendly policies. This enables them to establish a sound policy for managing natural resources and human capital. These results may affect the ecological efficiency of human development and bias our conclusions. We therefore exclude participants in these environmental conversions who score well.⁶ To complete

this part of our robustness checks, we exclude countries whose GDP per capita growth exceeds the average for the overall sample. This is justified by the fact that the causal effect of RBL on ecological efficiency may reflect the country's greater economic needs, not RBL. The results reported in Table A4, columns [28]–[34] show that despite the alternative samples, our results remain robust and consistent with the baseline results, reinforcing our confidence.

7.3. Combined entropy balancing and time perspective

Resource-backed loans have been on an upward trend over our period since the resource boom began in the early millennium (Mihalyi et al., 2020). Similarly, we can assume a significant improvement in life expectancy, education and income and respect for ecological overshoot over time in African countries, leading to an upward trend in ecological efficiency of human development (Hickel, 2020), without this being entirely explained by resource-backed loans. To account for the improvement in ecological efficiency over time and the increasing trend in the number of countries adopting resource-backed loans over our study period, we augment our base model by introducing a trend to redo the regressions. The new ATTs are shown in Table A5, columns [1]–[8] in Appendix. The new estimates remain similar to those of the baseline results (Table 3, columns [1]–[8]).

7.4. Placebo test

So far, our results reveal that RBLs have a statistically significant causal effect on ecological efficiency of human development in African countries. The effect considered in this paper may suffer from certain economic biases. In fact, the adoption of resource-backed loans can trigger a change in the economic environment, including increased investment, boosting GDP, increasing the share of educated, healthy and less poor population, which are Sustainable Development Goals, and civil unrest. In this sense, it can be argued that the effect observed may not be due to resource-backed loans, but to changes in institutional, political, social, or economic conditions after the adoption of these loans. Similarly, any other characteristic that may determine the adoption of resource-backed loans, but is not included in the econometric specification, may be a source of endogeneity even if we have added an additional variable battery for controls. We are aware that the non-transparency evoked in RBLs, according to the study of Mihalyi et al. (2020), may suggest that our control group may be contaminated. To circumvent these problems, it would be interesting to randomly assign a new treatment to the RBL variable. Indeed, it is expected that this new treatment, other than the actual dates, will not affect the ecological efficiency of human development. If we find a significant effect, this would mean that our results are affected by unobservable counterfactual and that our control group constructed from the information gathered by Mihalyi et al. (2020) is indeed contaminated. We report the ATT results in Table 7, columns [1]–[8]. We find no significant effect whatever the specifications, which clearly indicates that our results are not affected by an unobservable and that our base control group defined from the database of Mihalyi et al. (2020) does not suffer from any bias.

8. Transmission channels

This section empirically tests the main channels through which the causal effects of RBL can be exerted. Since the literature does not impose any explicit and singular methods for evaluating transmission channels, we adopt a simple two-step approach. First, we estimate a simple univariate regression of the potential channel on ecological efficiency of human development, using the OLS estimator. Second, we re-estimate our baseline model using a simple OLS estimator and replacing our dependent variable with the potential channel. In fact, the aim of this exercise is to test whether, for example, if stimulating economic activity

⁶ The highest score for this environmental convention is 1. A country with a score of 1 means that it is an active participant in environmental conventions.

Table 7

Resource-backed loans and sovereign debt default: Placebo test.

VARIABLES	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
RBL dummy (random)	0.0141 (0.0392)	−0.0007 (0.0284)	0.0267 (0.0329)	−0.0094 (0.0094)	−0.0227 (0.0405)	0.0025 (0.0174)	−0.0147 (0.0378)	−0.0053 (0.0071)
Constant	−0.6795*** (0.0127)	−0.7164*** (0.0440)	−0.7430*** (0.0668)	−0.7807*** (0.0366)	−0.5270*** (0.0324)	−0.7127*** (0.0483)	−0.5450*** (0.0631)	−0.6345*** (0.0477)
Covariates	No	No	No	No	Yes	Yes	Yes	Yes
Country/FE	No	Yes	No	Yes	No	Yes	No	Yes
Year/FE	No	No	Yes	Yes	No	No	Yes	Yes
Observations	611	611	611	611	611	611	611	611
R-squared	0.0006	0.7927	0.1636	0.9275	0.2769	0.8469	0.4098	0.9310

Notes: This Table presents the effect of resource-backed loans on ecological efficiency of human development obtained by entropy balancing method. The treatment variable is resource-backed loan (random assignment). The outcome variable is ecological efficiency of human development. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

through wealth creation positively affects ecological efficiency, and GDP per capita is a relevant channel through which RBL is effective, then GDP per capita should in turn increase ecological efficiency. In columns [1]–[3] of panel A (Table A6), we estimate a uni-variate regression of education level, income and expected years of schooling. All coefficients are positive and statistically significant at the 1% level. These results corroborate those found by [Hickel \(2020\)](#). Finally, the results reported in panel B show that resource-backed loans have a positive and significant impact on all three variables. These results reveal that the positive effect of resource-backed loans is mediated by an increase in the level of education, an improvement in expected years of schooling and income. In fact, resource-backed loans will increase public investment in infrastructure (health, education, and roads), boosting GDP, increasing the share of educated, healthy and less poor population.

9. Heterogeneity

9.1. Effect by type of resource-backed loans

In this section, we desegregated resource-backed loans into three types of resource-backed loans based on information from [Mihalyi et al. \(2020\)](#) database. The three types of loans signed by countries include oil-backed loans, mineral-backed loans, and cocoa and tobacco-backed loans. In fact, oil-backed loans are simply loans that must be repaid through the exploitation and sale of oil. Mineral-backed loans are those whose repayment is directly linked to bauxite, copper, cobalt, platinum, gold, and diamond, while cocoa and tobacco-backed loans are repaid by the exploitation and sale of cocoa and tobacco. These different forms of loan can have a heterogeneous effect on the ecological efficiency of human development in African countries. This will be discussed in this sub-section.

The results reported in Table 8, columns [1]–[3] show that only cocoa and tobacco-backed loans have a negative impact on ecological

Table 8

Heterogeneity by type of resource-backed loans.

VARIABLES	[1]	[2]	[3]
Cocoa and tobacco-backed loans	−0.0285*** (0.0100)		
Minerals resource-backed loans		0.0068* (0.0098)	
Oil resource-backed loans			0.0099** (0.0060)
Constant	−0.6753*** (0.0444)	−0.6759*** (0.0446)	−0.6788*** (0.0446)
Covariates	Yes	Yes	Yes
Country/FE	Yes	Yes	Yes
Year/FE	Yes	Yes	Yes
Observations	799	799	799
R-squared	0.9395	0.9393	0.9395

Notes: This Table presents the results of desegregated resource-backed loans obtained by entropy balancing method. Standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

efficiency of human development. In contrast, oil-backed and mineral-backed loans have a positive and significant impact on the outcome variable. Indeed, the negative impact of cocoa- and tobacco-backed loans could be explained by the fact that cocoa and tobacco production is not sustainable and does not meet environmental standards in terms of sustainable development in African countries. In addition, these results are in perfect harmony with those found by [Ruf et al. \(2015\)](#) that cocoa farming is one of the main drivers of deforestation in Africa.

While our results indicate that mineral and oil-backed loans have a positive impact on the ecological efficiency of humans, they can also be the cause of significant environmental degradation. Indeed, according to [Coulibaly \(2023\)](#), mineral-backed loans exert strong pressure on forest cover. The author explains the positive effect of mineral-backed loans on deforestation by the instability of commodity prices, which increases the value of the debt to be repaid, given that the loans are signed in US dollars. This will lead to overexploitation and depletion of natural resources, affecting forest cover. However, the key idea of the Sustainable Development Index designed by [Hickel \(2020\)](#) is to incorporate the ecological dimension, i.e. CO2 emissions and material borrowings that do not exceed a well-defined planetary limit. If this is the case, a related question in this paper would be to analyze the scale of resource-backed loans on ecological impact and planetary boundaries, in order to draw solid recommendations for individual countries.

9.2. Effect by public debt level

In this subsection, we attempt to capture resource-backed loans by level of public debt. Our aim is to capture the level of debt that would be compatible with ecological efficiency of human development for countries adopting RBL. Indeed, one method that is well-suited to this type of exercise is the use of threshold models ([Granovetter, 1978](#)). However, the unbalanced format of our panel data means that this technique cannot be rigorously applied. To this end, we used the entropy balancing method, setting conditions on public debt. In other words, we look at the effect of resource-backed loans on the ecological efficiency of human development when the level of public debt is below the sample average and when it is above the average. The idea is that a high public debt could exacerbating the country's public debt situation and reduce financing capacity for clean, favorable, and sustainable development. We will test this hypothesis.

The results presented in Table 9, column [1] indicate that resource-backed loans have a positive and statistically significant impact on ecological efficiency of human development when the level of public debt is relatively low, while no effect is found when the level of indebtedness exceeds the average for the overall sample (Table 9, column [2]). These results are not surprising, as unsustainable public debt reduces budgetary space and prevents the government from taking action to protect the environment and biodiversity. Countries should therefore avoid taking on too much debt at the risk of losing fiscal discipline and should not use resource-backed loans. These results corroborate those found by [Boly et al. \(2022\)](#) according to which public

Table 9
Heterogeneity by debt level.

VARIABLES	[1]	[2]
	Public Debt < mean	Public Debt > mean
RBL dummy	0.0463*** (0.0092)	0.0004 (0.0148)
Constant	−0.6147*** (0.0391)	−0.8410*** (0.0442)
Covariates	Yes	Yes
Country/FE	Yes	Yes
Year/FE	Yes	Yes
Observations	509	290
R-squared	0.9721	0.9764

Robust standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

debt and environmental debt are complementary in the long term and generally substitute each other in the short term. This means that in the long term, public debt offsets environmental debt, so that public debt has an overall negative effect on pollution control spending; whereas in the short term, the second effect outweighs the first, provided that public debt is not too high.

9.3. Does ecological impact and planetary boundaries matter?

The analysis of transmission channels in [section 8](#) shows that resource-backed loans have a positive effect on the GNI of signatory countries. Indeed, analysis of the 2015 HDI data clearly shows that some countries are approaching an achievable state while achieving very high levels (0.8 or more) on the life expectancy index with as little as \$3300 per capita (or 0.9 with as little as \$11,000), and very high levels on the education index with as little as \$8700 per capita, while maintaining ecological impact at sustainable levels ([Hickel, 2020](#)). In addition, another countries are achieving fairly high levels of sustainable development, with per capita figures of \$10,000 to \$14,000 ([O'Neill et al., 2018](#); [Hickel, 2019](#)). However, when GNI exceeds this level, it begins to have clear negative social and ecological consequences ([Inglehart, 2020](#); [Lamb et al., 2014](#); [Kubiszewski et al., 2013](#); [Deaton, 2008](#); [Max-Neef, 1995](#)).

Our concern here is that if resource-backed loans help boost GNI beyond ecologically compatible levels, they could adversely affect environmental sustainability. We are testing this hypothesis to ensure that resource-backed loans do not affect the environment and achieve an achievable planetary limit. To achieve this, we have calculated a composite⁷ environmental sustainability index that captures CO₂ and greenhouse gas emissions, the key determinants of ecological impact ([Han and Chatterjee, 1997](#); [Karakurt and Aydin, 2023](#); [Jiang et al., 2019](#); etc.). The results presented in [Table A7](#); Panel A confirm our hypothesis that resource-backed loans contribute to increase ecological impact in signatory countries compared to non-signatory countries. Further analysis in Panels B and C shows that the impact is stronger on CO₂ emissions than on greenhouse gases ([Table A7](#), column [8]). One possible explanation for this result is that RBLs, by depleting natural resources and destroying forest cover ([Coulibaly, 2023](#)), accelerate CO₂ and greenhouse gas emissions, since forests sequester carbon and protect the ozone layer. These results reveal that the environmental concerns of RBLs are significant in the context of the sustainable development and energy transition so sought-after around the world and will need to be brought to the attention and information of policy-makers in resource-rich countries.

⁷ The index is calculated using the [Anderson \(2008\)](#) method.

9.4. Does implementing the extractive industries transparency initiative (EITI) matter?

In this subsection, we examine the effectiveness of resource-backed loans in the presence of the Extractive Industries Transparency Initiative (EITI). Indeed, the EITI specifically encourages transparency, particularly with regard to RBLs ([Mihalyi et al., 2020](#)) and prevents poor governance of natural resources. For instance, [Kinda and Thiombiano \(2024\)](#) using entropy balancing methods on a sample of 83 resource-rich developing countries from 2001 to 2017, find that EITI implementation significantly reduces forest cover loss by around 300 to 760 ha.

These results confirm that the implementation of EITI is an effective policy program for limiting the negative environmental impacts caused in part by the extractive industries. With respect to resource-backed loans, countries that have signed an RBL and are EITI members were required to comply with the 2019 EITI standard, which includes a transparency requirement related to payments associated with RBLs as part of the payment flows that EITI-implementing countries must disclose. Other requirements relate to the disclosure of barter transactions in which oil or mining products are exchanged for goods or services, including infrastructure or even loans, not only for countries but also for companies involved in the exploitation of natural resources ([Mihalyi et al., 2020](#)). In contrast, [Mawejje \(2019\)](#) believes that EITI is not a panacea for eliminating the tax revenue curse in resource-rich African countries. Using panel fixed-effects econometrics and dynamic panel GMM on a panel of 31 resource-rich sub-Saharan African countries, the author shows that EITI membership has had a small positive effect on non-oil revenue mobilization, and only partially offsets the negative impact of resource dependence on non-resource tax revenues. These results suggest that, although improving governance and transparency in the management of natural resource revenues through EITI is seen as essential to mitigating the negative consequences that characterize resource-rich countries, its impact is not sufficiently clear.

This paper therefore tests whether EITI membership can still help to improve the ecological efficiency of human development for RBL signatory countries. To test this hypothesis, we include an interaction between each of the EITI and RBL variables. Consequently, a statistic significant interaction means the presence of heterogeneity and the effectiveness of resource-backed loans in the presence of EITI membership.

The results presented in [Table 10](#), columns [1]–[3] support our hypothesis even if the amplitudes of the coefficients remain quite similar to the baseline results (slight improvement). Indeed, the positive and statistically significant sign of the interaction between RBL and each of the EITI variables, i.e., EITI full, EITI commitment, and EITI candidate, suggests that the effect of RBL may be amplified in situations of EITI implementation, commitment, and EITI candidate. These results corroborated with those found by [Kinda and Thiombiano \(2024\)](#).

Table 10
Effect of EITI implementing.

VARIABLES	[1]	[2]	[3]
	Ecological efficiency of human development		
RBL*EITI countries	0.0211*** (0.0068)		
RBL*EITI commitment		0.0211*** (0.0068)	
RBL*EITI candidate			0.0257*** (0.0066)
Constant	−0.7190*** (0.0276)	−0.7190*** (0.0276)	−0.7212*** (0.0276)
Covariates	Yes	Yes	Yes
Country/FE	Yes	Yes	Yes
Year/FE	Yes	Yes	Yes
Observations	602	602	602
R-squared	0.9707	0.9707	0.9711

Robust standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

9.5. Do financial market access conditions matter?

Given that the financing of sustainable development for African countries requires the mobilization of resources on the financial markets, we link resource-backed loans to bond market interest rates.⁸ In other words, we test whether resource-backed loans replace or complement access to credit. The idea of understanding whether resource-backed loans and access to financial markets are complementary or substitutable answers two arguments. The first is that if a resource-rich country with high financial risk limiting its access to financial markets uses its natural resource endowment as collateral, lenders may perceive this behavior as a risky debtor. In fact, they can say because the country has a fragile financial situation, and it is likely that it will not be able to repay its debt and therefore it uses its natural resources as collateral. As a result, resource-backed loans will replace access to credit since traditional lenders will either refuse to lend or will lend at high interest rates which will ultimately discourage debtors and they will move towards the resource-backed loan market. The second argument is that donors may perceive the resources as a good guarantee and think that whatever the situation prevails, the debtor country will be able to repay since there is a guarantee. From this point of view, resource-backed loans will complement other loans and strengthen access to financial markets.

To test this, we use two indicators of access to financial markets. Firstly, we use the interest rate on the bond market as the first instrument that conditions the evolution of the bond market. Second, following Sawadogo (2020), we use credit rating as an indicator of access to financial markets. The results reported in Table 11, columns [1]–[2] show that resource-backed loans have a positive and statistically significant impact on the bond market interest rate. Naturally, a higher interest rate limits the ability of debtor countries to access the financial markets. By evaluating its impact on credit rating, our results show that resource-backed loans have a negative and statistically significant impact (Table 11, columns [3]–[4]). Which suggests that rather than being a complement to ordinary loans, resource-backed loans would replace them.

9.6. The contribution of structural factors

We explore the important structural differences that shape the effect of resource-backed loans on ecological efficiency of human development in African countries despite a number of common characteristics. To achieve this, we apply the control function regression approach of Lin and Ye (2009). Using this method, we investigate the sensitivity of our different baseline results to certain country structural characteristics,

Table 11
Resource-backed loans and credit access.

VARIABLES	[1] Interest rate	[2] Interest rate	[3] Credit rating	[4] Credit rating
Resource-backed loans	0.5723*** (0.1560)	1.1318*** (0.3196)	−1.4910*** (0.3318)	−0.9115** (0.4205)
Constant	2.0051*** (0.0578)	1.8003*** (0.6462)	8.4003*** (0.1998)	8.6407*** (1.0496)
Country/FE & Year/ FE	No	Yes	No	Yes
Covariates	No	Yes	No	Yes
Observations	538	538	302	302
R-squared	0.0913	0.6601	0.1062	0.9118

Robust standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

⁸ We thank an anonymous reviewer for suggesting this idea.

such as: age dependency ratio, financial development, government revenue, interest on debt, urban population, government effectiveness, credit rating, public debt, and resource depletion. To do that, we include an interaction between each of these variables and resource-backed loans. Consequently, a statistically significant interaction means the presence of heterogeneity. The Table A8 in the Appendix presents the estimate of the treatment effect of resource-backed loans based on the control function regression approach. In the first column, we perform a simple OLS regression of resource-backed loans on an ecological efficiency of human development variable, including the estimated propensity score (Pscore) obtained from our baseline probit model as the control function. The estimated coefficients, which capture the difference between the mean resource-backed loans of beneficiaries and non-beneficiaries, are positive and statistically significant. A statistically significant coefficient on the propensity score means that there is self-selectivity in the model. However, the coefficient is not significant even when certain structural factors are controlled (columns [2]–[10]). The results indicate that resource-backed loans are more effective in increasing ecological efficiency of human development in countries that have them, particularly when: (i)- the government is more effectiveness, (ii)- interest on debt is low, (iii)- there is a small urban population, (iv)- fiscal discipline is captured by low debt levels, (v)- natural resources are abundant. More specifically, the effect of resource-backed loans on the ecological efficiency of human development is greater in countries where natural resource depletion is lower.

10. Conclusion

This paper analyzes the causal effect of resource-backed loans on the Sustainable Development Index measured by ecological efficiency of human development using a large sample of 45 African countries over the period 1990–2019. To achieve this, we began by exploring the correlation between the two variables through a graphical analysis of the stylized facts. Although we obtained graphical evidence that RBLs are likely to go hand in hand with the ecological efficiency of human development, we have deepened our analysis by determining the causal effect of these RBLs through the use of the entropy balancing method. We show that irrespective of the specifications, resource-backed loans adoption increases ecological efficiency of human development in resource-backed loans countries compared to non-resource-backed loans countries. The magnitude of the coefficients varies between 0.024 percentage points (column [8]) and 0.163 percentage points (column [2]) of Table 3. This result is robust given the relative stability of the coefficients across the eight specifications in the table, with an average effect of 0.076 percentage points. In other words, resource-backed loans adoption increases ecological efficiency of human development by 0.076 percentage points in resource-backed loans countries compared to non-resource-backed loans. These results are robust to various tests, including alternative specifications and alternative estimation methods. The analysis of transmission channels indicates that the expected level of education, income and years of schooling drive the stabilizing effect of resource-backed loans. However, the results reveal some heterogeneity in the type of resource-backed loans, in the burden of public debt and structural factors such as interest on debt, urban population, government effectiveness, public debt, and resource depletion. Additional results highlighted in this study show that resource-backed loans contribute to CO2 and greenhouse gas emissions in African countries. This result is not surprising, as it has been shown in the literature that financing real income growth through external public debt can lead to increased CO2 emissions and environmental degradation (e.g., Boly et al., 2022; Katircioglu and Celebi, 2018). This result should attract the attention of policy-makers in countries wishing to sign a resource-backed loan agreement.

In addition, it appears that the positive impact is conditioned by the level of sovereign debt. Finally, EITI membership has a stabilizing effect on the positive impact of resource-backed loans. This result

corroborates those found in the literature on the beneficial effects of EITI membership (see, for instance, Kinda and Thiombiano, 2024; Villar, 2022; Villar and Papyrakis, 2017). In the literature, this study contributes, on the basis of a cross-national analysis, to the debate on the role of new sources of sustainable development financing on well-being and environmental sustainability in African countries.

In sum, we present two aspects of the risk of leveraged loans beyond their benefits, namely debt risk and environmental risk. Our results point to the existence of an inverted U-shaped Environmental Kuznets Curve (EKC) between economic development and the environmental performance of African economies. It would therefore be convenient for resource-backed loans, particularly from cocoa and tobacco, to include biodiversity-related elements and strict provisions for the implementation of actions to mitigate their adverse impact on the environment. Policy-makers need to heed these observations in order to take advantage of resource-backed loans, circumvent the natural resource curse and ensure cleaner, more inclusive sustainable development over the long term. Consequently, EITI membership would be a best policy choice to achieve these goals, as the initiative's standards are based on good governance, transparency, and environmental compliance. When it comes to sovereign debt management, the leaders of African countries need to work towards a climate finance goal, i.e. adopt a long-term plan to protect nature and biodiversity; and to do this they need to engage in debt consolidation programs (e.g. by implementing tax increases, public spending cuts or debt relief agreements). In fact, debt servicing reduces the government's ability to finance emissions-reduction spending, which degrades environmental quality and increases long-term climate debt. By reducing public debt, therefore, the government provides more fiscal space to finance nature protection policies in the long term.

Finally, one of the main limitations of this paper lies in the use of a binary variable to capture resource-backed loans, rather than using

actual data on these resource-backed loans. Future research could aim to correct this by using data on resource-backed loans. In addition, the paper says nothing about the effect of resource-backed loans on the loss of forest cover, which is a good indicator of ecology and environmental quality. We intend to explore this avenue in the future.

CRedit authorship contribution statement

Yacouba Coulibaly: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Robustness, descriptive statistics, and source of data

Table A1

Resource-backed loans and ecological efficiency: using OLS.

VARIABLES	[1] OLS	[2] OLS	[3] OLS	[4] OLS	[5] OLS	[6] OLS
RBL dummy	0.0043*** (0.0109)	0.0349*** (0.0101)	0.0400*** (0.0105)	0.0331*** (0.0110)	0.0315*** (0.0109)	0.0322*** (0.0109)
Log Inflation		0.0023 (0.0029)	0.0008 (0.0031)	0.0024 (0.0035)	0.0018 (0.0035)	0.0019 (0.0035)
Log ODA			0.0122** (0.0056)	0.0106* (0.0062)	0.0088 (0.0062)	0.0086 (0.0062)
Log FDI				0.0049* (0.0025)	0.0050* (0.0025)	0.0048* (0.0026)
Log Natural resources rents					0.0185** (0.0090)	0.0162* (0.0097)
Log Resource depletion						0.0023 (0.0027)
Constant	−0.8170*** (0.0281)	−0.8110*** (0.0367)	−0.8223*** (0.0398)	−0.8692*** (0.0490)	−0.9298*** (0.0601)	−0.9289*** (0.0604)
Country/FE & Year/FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1066	881	866	799	799	799
R-squared	0.8518	0.8477	0.8482	0.8491	0.8502	0.8503

Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2

Resource-backed loans and ecological efficiency: adding control variables by groups.

	[1]	[2]	[3]	[4]	[5]
VARIABLES	Economic	Political and institutional	Climate	Infrastructure and structural	All groups
	variables group	conditions group	considerations group	issues group	
RBL dummy	0.0241*** (0.0073)	0.0180** (0.0073)	0.0240*** (0.0071)	0.0252*** (0.0070)	0.0187*** (0.0072)
Constant	−0.6971*** (0.0455)	−0.6832*** (0.0447)	−0.6959*** (0.0459)	−0.6464*** (0.0479)	−0.6442*** (0.0489)
Covariates	Yes	Yes	Yes	Yes	Yes
Country/FE	Yes	Yes	Yes	Yes	Yes
Year/FE	Yes	Yes	Yes	Yes	Yes
Observations	799	755	796	773	745
R-squared	0.9403	0.9475	0.9398	0.9476	0.9483

Standard errors are in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01.

Table A3

Resource-backed loans and ecological efficiency: adding control variables desegregated.

	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]
VARIABLES	GDP volatility	Investment volatility	Total investment	Stock price volatility	Financial openness	Rich-resources	Trade openness	Remittance inflows	Natural shock index	Public debt
RBL dummy	0.0341*** (0.0073)	0.0291*** (0.0070)	0.0216*** (0.0071)	0.2578*** (0.0242)	0.0227*** (0.0075)	0.0241*** (0.0072)	0.0277*** (0.0079)	0.0116* (0.0066)	0.0261*** (0.0072)	0.0345*** (0.0071)
Constant	−0.7076*** (0.0472)	−0.7244*** (0.0449)	−0.7090*** (0.0485)	−0.9352*** (0.0494)	−0.6665*** (0.0471)	−0.7055*** (0.0413)	−0.6950*** (0.0524)	−0.6314*** (0.0483)	−0.7202*** (0.0487)	−0.6129*** (0.0361)
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	738	716	775	117	780	799	709	714	762	670
R-squared	0.9519	0.9517	0.9409	0.9685	0.9383	0.9401	0.9402	0.9456	0.9421	0.9610

	[11]	[12]	[13]	[14]	[15]	[16]	[17]	[18]	[19]	[20]
VARIABLES	Rule of law	Corruption	External conflict	Internal conflict	Political stability	Climatic vulnerability	Disasters index	Exposure index	Shock index	Exchange rate
RBL dummy	0.0239*** (0.0072)	0.0191** (0.0084)	0.0163** (0.0083)	0.0184** (0.0082)	0.0222*** (0.0074)	0.0462*** (0.0073)	0.0228*** (0.0071)	0.0223*** (0.0071)	0.0250*** (0.0070)	0.0209** (0.0084)
Constant	−0.8583*** (0.0478)	−0.6583*** (0.0474)	−0.7968*** (0.0614)	−0.6927*** (0.0535)	−0.7746*** (0.0495)	−1.4236*** (0.1711)	−0.6810*** (0.0470)	−0.3407*** (0.0873)	−0.7081*** (0.0452)	−0.6550*** (0.0453)
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	706	553	553	553	706	638	774	760	762	708
R-squared	0.9550	0.9449	0.9482	0.9451	0.9517	0.9524	0.9401	0.9512	0.9420	0.9362

Standard errors are in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01.

Table A4

Resource-backed loans and ecological efficiency: adding control variables desegregated (continued) and modification sample.

	[21]	[22]	[23]	[24]	[25]	[26]	[27]
VARIABLES	ICT infrastructure	Education	Innovation	Doing business	Electric power consumption	Access to electricity urban	Access to electricity rural
RBL dummy	0.0357*** (0.0067)	0.0379*** (0.0073)	0.0432** (0.0173)	0.0425*** (0.0079)	0.0157* (0.0083)	0.0287*** (0.0071)	0.0286*** (0.0067)
Constant	−0.4899*** (0.0531)	−0.7470*** (0.0527)	−0.6848*** (0.0654)	−0.7573*** (0.0542)	−0.5192*** (0.0435)	−0.7670*** (0.0334)	−0.6236*** (0.0493)
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	638	638	393	638	367	731	627
R-squared	0.9610	0.9535	0.8991	0.9507	0.9635	0.9516	0.9594

	[28]	[29]	[30]	[31]	[32]	[33]	[34]
Excluding	Hyperinflation episode	New RBL	Pre-2004	Financial crisis year	Environmental convention	Global demand	Oil crisis year
RBL dummy	0.0171**	0.0167**	0.0216***	0.0229***	0.0224***	0.0335**	0.0237***

(continued on next page)

Table A4 (continued)

	[28]	[29]	[30]	[31]	[32]	[33]	[34]
Excluding							
	Hyperinflation episode	New RBL	Pre-2004	Financial crisis year	Environmental convention	Global demand	Oil crisis year
Constant	(0.0073) −0.6619*** (0.0469)	(0.0078) −0.6495*** (0.0451)	(0.0063) −0.6879*** (0.0269)	(0.0079) −0.7010*** (0.0479)	(0.0083) −0.6814*** (0.0471)	(0.0146) −0.7566*** (0.0571)	(0.0071) −0.7050*** (0.0461)
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	773	708	524	729	686	333	768
R-squared	0.9415	0.9360	0.9788	0.9372	0.9319	0.9651	0.9397

Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5

Resource-backed loans and ecological efficiency: perspective time.

VARIABLES	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
RBL dummy	0.1165*** (0.0209)	0.0379*** (0.0107)	0.0611*** (0.0206)	0.0286*** (0.0080)	0.0899*** (0.0169)	0.0326*** (0.0092)	0.0443*** (0.0167)	0.0241*** (0.0072)
Constant	−0.7441*** (0.0210)	−0.9018*** (0.0201)	−0.7784*** (0.0725)	−0.7837*** (0.0402)	−0.6446*** (0.0279)	−0.8200*** (0.0308)	−0.6301*** (0.0685)	−0.7031*** (0.0465)
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Trend	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	799	799	799	799	799	799	799	799
R-squared	0.1216	0.9299	0.2255	0.9382	0.4606	0.9327	0.5275	0.9401

Notes: This Table presents the causal effect of RBL adoption on ecological efficiency of human development obtained by entropy balancing method introducing the trend. Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix B. Steps of Anderson's synthetic index

The six steps described by [Anderson \(2008\)](#) to calculate this synthetic index are as follows:

- 1- Select k indicators relevant for outcome j .
- 2- Adjust sign: For all k indicators, ensure the positive direction always indicates a “better outcome”.
- 3- Normalize indicators: Demean all k indicators by subtracting the mean of the indicator in the reference group (full sample is the default reference group). Then, convert them to effect sizes, $\tilde{Y}_k$ by dividing each indicator by its reference group standard deviation.⁹

Construct weights: Create weights using Σ^{-1} , the inverse of the covariance matrix of the normalized indicators.¹⁰ Specifically, set the weight $\tilde{w}_k$ on each indicator equal to the sum of its row entries in Σ^{-1} . With this rule, highly correlated indicators are assigned small or offsetting weights, while less correlated outcomes receive larger weights.

- 4- Construct index: Calculate the weighted average of $\tilde{Y}_k$ for observation i . Formally, the weighted average $\bar{S}_i$ is calculated using $\bar{S}_i = (\mathbf{1}' \Sigma^{-1} \mathbf{1})^{-1} (\mathbf{1}' \Sigma^{-1} \tilde{Y}_i)$ where $\mathbf{1}$ is a column vector of 1s and $\tilde{Y}_i$ is a column vector of all outcomes for observation i . This is an efficient GLS estimator.
- 5- Normalize index: Demean index $\bar{S}_i$ by subtracting the mean of the index in the reference group, and convert it to effect sizes by dividing it by its reference group standard deviation.¹¹

Table A6

Validity of transmission channels.

Panel A	[1]	[2]	[3]
VARIABLES	Ecological efficiency	Ecological efficiency	Ecological efficiency
Education level	0.0555*** (0.0095)		
GNI per capita		0.1221*** (0.0220)	

(continued on next page)

⁹ Note that in some cases, particularly when all subindicators are of comparable scales, the normalization may not be necessary. The `swindex` command accommodates this special case using the `nostd` option (see, [Schwab et al. \(2020\)](#) on the command).

¹⁰ This is equivalent to using the correlation matrix in the special case that the normalization procedure described in step 3 standardizes against the full sample using the default option.

¹¹ This step is implicitly done by [Anderson \(2008\)](#), although it is not formally discussed. This step is optional using the `swindex` command with the `norescale` option. The default uses the reference group from step 3, but the full sample mean, and standard deviation can alternatively be used to normalize the index with the `fullrescale` option.

Table A6 (continued)

Panel A	[1]	[2]	[3]
VARIABLES	Ecological efficiency	Ecological efficiency	Ecological efficiency
Expected years of schooling			0.4484*** (0.0206)
Constant	−0.6282*** (0.0618)	−2.0011*** (0.2014)	−1.6067*** (0.0479)
Covariates	Yes	Yes	Yes
Country/FE & Year/FE	Yes	Yes	Yes
Observations	638	799	799
R-squared	0.8776	0.8556	0.9089

Panel B	[1]	[2]	[3]
	Education level	GNI per capita	Expected Years of Schooling
RBL dummy	0.4470*** (0.0762)	0.0505* (0.0278)	0.0608** (0.0250)
Constant	−5.1386*** (0.1970)	8.8337*** (0.0762)	1.4381*** (0.0686)
Covariates	Yes	Yes	Yes
Country/FE & Year/FE	Yes	Yes	Yes
Observations	731	854	854
R-squared	0.9048	0.9738	0.8786

Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7

Resource-backed loans and environmental sustainability.

VARIABLES	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
Panel A	Ecological impact and planetary boundaries							
RBL dummy	0.2766*** (0.1044)	0.4036*** (0.0316)	0.1864* (0.1035)	0.0866*** (0.0324)	0.2766*** (0.0644)	0.2744*** (0.0424)	0.2617*** (0.0734)	0.1052*** (0.0314)
Constant	0.0787* (0.0462)	0.7746*** (0.0375)	−0.1728 (0.4989)	0.5927 (0.3720)	0.2946** (0.1273)	0.8130*** (0.1516)	0.2828 (0.3070)	0.4328 (0.3993)
Covariates	No	No	No	No	Yes	Yes	Yes	Yes
Country/FE	No	Yes	No	Yes	No	Yes	No	Yes
Year/FE	No	No	Yes	Yes	No	No	Yes	Yes
Observations	926	926	926	926	926	926	926	926
R-squared	0.0245	0.9247	0.0652	0.9567	0.5473	0.9483	0.5640	0.9643

Panel B	CO2 emissions							
RBL dummy	0.1034 (0.1162)	0.5201*** (0.0537)	−0.0278 (0.1226)	0.1669*** (0.0526)	0.0670 (0.0932)	0.3945*** (0.0713)	0.0286 (0.0983)	0.1358** (0.0538)
Constant	−0.9411*** (0.0621)	−0.6675*** (0.0729)	−0.3587 (0.3308)	−0.2733* (0.1475)	0.7081*** (0.1292)	−0.4122* (0.2271)	1.7955*** (0.4111)	−0.2444 (0.2643)
Covariates	No	No	No	No	Yes	Yes	Yes	Yes
Country/FE	No	Yes	No	Yes	No	Yes	No	Yes
Year/FE	No	No	Yes	Yes	No	No	Yes	Yes
Observations	853	853	853	853	853	853	853	853
R-squared	0.0023	0.9016	0.0491	0.9304	0.4967	0.9112	0.5434	0.9337

Panel C	Greenhouse emissions							
RBL dummy	0.2035 (0.2298)	0.3792*** (0.0278)	0.2021 (0.2281)	0.0549* (0.0298)	0.2035 (0.1659)	0.2253*** (0.0427)	0.2030 (0.1765)	0.0696** (0.0304)
Constant	9.6584*** (0.0729)	10.9164*** (0.0314)	8.9467*** (0.6937)	10.4880*** (0.4425)	8.6223*** (0.2757)	10.8984*** (0.1677)	7.9441*** (0.4807)	10.2625*** (0.4836)
Covariates	No	No	No	No	Yes	Yes	Yes	Yes
Country/FE	No	Yes	No	Yes	No	Yes	No	Yes
Year/FE	No	No	Yes	Yes	No	No	Yes	Yes
Observations	926	926	926	926	926	926	926	926
R-squared	0.0035	0.9786	0.0248	0.9884	0.4032	0.9860	0.4189	0.9905

Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8

Resource-backed loans and ecological efficiency of human development: structural factors.

VARIABLES	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]
RBL dummy	0.0305*** (0.0100)	0.0636 (0.0610)	0.0422** (0.0196)	0.0448** (0.0190)	0.0476*** (0.0149)	0.1437*** (0.0206)	0.0225* (0.0119)	−0.0055 (0.0468)	0.0770*** (0.0138)	0.0504*** (0.0137)
Pscore	−0.0437 (0.0375)	−0.0449 (0.0378)	−0.0456 (0.0376)	0.1262 (0.0443)	−0.1163 (0.0450)	−0.0277 (0.0362)	0.0028 (0.0337)	0.1470 (0.0777)	−0.0034 (0.0340)	−0.0325 (0.0371)
RBL*Age dependency ratio		−0.0060 (0.0099)								
RBL*Financial development			−0.1104 (0.1590)							
RBL*Government revenue				−0.0010 (0.0008)						
RBL*Interest on debt					−0.0125*** (0.0037)					
RBL*Urban population						−0.0023*** (0.0004)				
RBL*Government effectiveness							0.0159* (0.0089)			
RBL*Credit rating								0.0043 (0.0055)		
RBL*Public debt									−0.0007*** (0.0002)	
RBL* Resource depletion										−0.0014** (0.0007)
Constant	−0.8289*** (0.0413)	−0.8299*** (0.0414)	−0.8269*** (0.0414)	−0.7146*** (0.0317)	−0.7248*** (0.0320)	−0.8213*** (0.0427)	−0.8050*** (0.0376)	−0.5706*** (0.0264)	−0.7150*** (0.0384)	−0.8237*** (0.0420)
Country/FE & Year/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	799	799	792	483	483	799	645	304	670	799
R-squared	0.8477	0.8477	0.8469	0.8820	0.8824	0.8492	0.8867	0.9216	0.8913	0.8480

Standard errors are in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01.

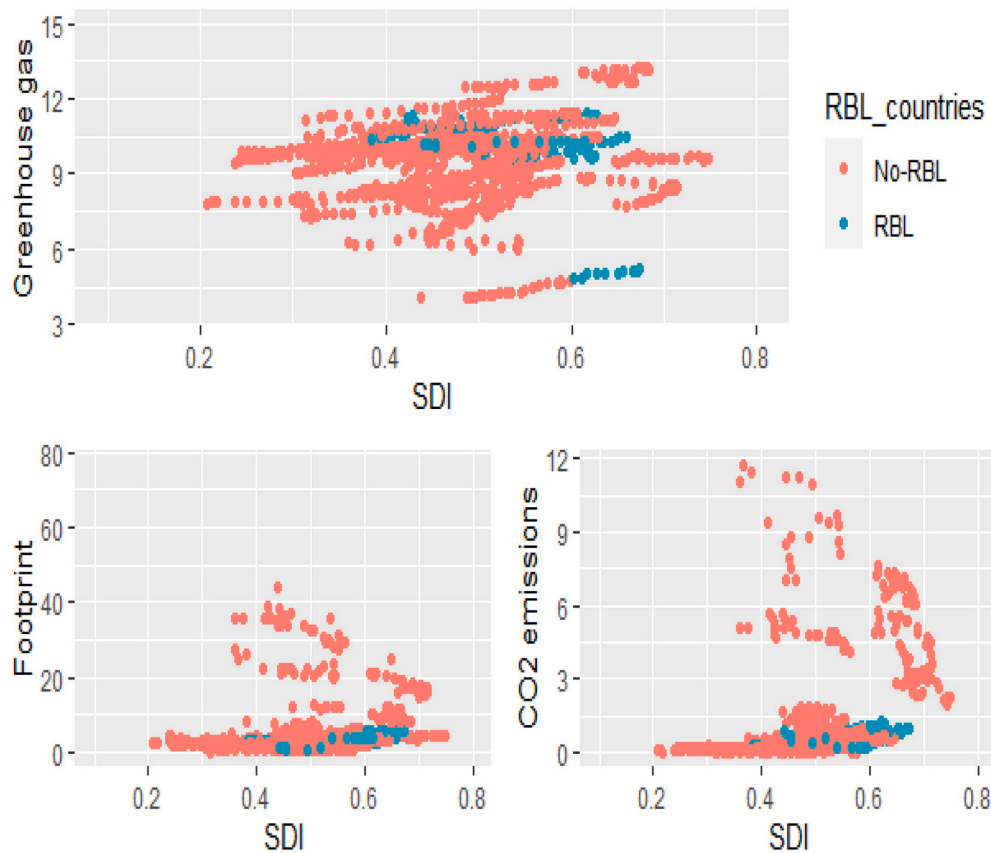


Fig. 3. Correlation between SDI, greenhouse gas, CO2 emissions, and footprint.

Note: This graph is the authors' own elaboration, using data from [Mihalyi et al. \(2020\)](#) and [Hickel \(2020\)](#). Scatterplot of SDI, greenhouse gas emissions, CO2 emissions and material footprint (1990–2019).

Table A9

Details on resource-backed loans in African countries.

Country	Loan year	Loans in millions of USD	Resources used	loans country	loans entity	Sectors targeted for investment
Angola	2004	2000	Oil	China	Eximbank	Infrastructure
Angola	2007	500	Oil	China	Eximbank	Infrastructure
Angola	2007	2000	Oil	China	Eximbank	Infrastructure
Angola	2009	2000	Oil	China	Eximbank	Infrastructure
Angola	2010	2500	Oil	China	ICBC	housing
Angola	2015	15,000	Oil	China	CDB	Infrastructure
Chad	2013	600	Oil	International	Glencore	Budget support and debt refinancing
Chad	2014	1356	Oil	International	Glencore	Oil
RDC	2008	3000	Copper & Cobalt	China	Eximbank	Infrastructure
RDC	2011	500	Copper	Corée	Korea Exim	Infrastructure
Ghana	2011	1500	Oil	China	CDB	Infrastructure
Ghana	2011	1500*	Oil	China	CDB	Infrastructure
Ghana	2018	2000	Bauxite	China	Sinohydro	Infrastructure
Guinea	2017	20,000	Bauxite	China	CCC	Infrastructure
Niger	2013	1000*	Oil	China	Eximbank	Oil
Republic of Congo	2006	1600	Oil	China	Eximbank	Infrastructure
Republic of Congo	2011	625	Oil	International	Gunvor	Oil
Republic of Congo	2012	1000	Oil	China	Eximbank	Infrastructure
Republic of Congo	2015	1000	Oil	International	Trafigura	Unknown
Republic of Congo	2015	850	Oil	International	Glencore	Unknown
Sao Tome and Principe	2010	30	Oil	Nigeria	Government	Oil
South Sudan	2015	75	Oil	International	CNPC	Unknown
South Sudan	2015	1000	Oil	China	Eximbank	Budget support
South Sudan	2016	169	Oil	China	Eximbank	Road
Sudan	2007	3000	Oil	China	Eximbank	Infrastructure
Zimbabwe	2006	200	platinum	China	Eximbank	Agricultural
Zimbabwe	2011	98	Diamond	China	Eximbank	Education

Source: Author's construction based on information from [Mihalyi et al. \(2022\)](#). Note: RBLs marked with * were subsequently cancelled without disbursement.

Table A10
Summary statistics for all variables.

Variable	Obs	Mean	Std. Dev.	Min	Max
SDI	1066	0.482	0.102	0.207	0.748
RBL 01	1350	0.086	0.28	0	1
RBL Oil	1350	0.012	0.108	0	1
RBL Mineral	1350	0.004	0.067	0	1
RBL Other	1350	0.001	0.038	0	1
Inflation rate	1048	1.895	1.37	−3.305	10.076
ODA	1260	1.778	1.289	−5.651	4.553
FDI	1170	0.433	1.828	−13.184	5.087
Natural resource rents	1267	1.859	1.574	−6.749	4.139
Resource depletion	1240	1.021	2.378	−8.822	4.634
GDP per capita volatility	1093	228.025	486.742	4.063	5387.647
Public Investment volatility	1032	4.148	4.618	0	32.959
Total investment	1196	22.664	12.622	2.663	115.102
Stock price volatility	125	13.12	6.81	1.973	34.384
Financial development	1290	0.118	0.094	0	0.643
Resources-rich	720	0.375	0.484	0	1
Natural shock index	1259	43.914	19.251	0	95.767
Public debt	1004	66.47	65.124	0	600.117
Rule of law	1075	0.365	0.127	0	0.721
Corruption	870	2.227	0.939	0	5
Internal conflict	870	7.854	2.149	0	12
External conflict	870	9.317	1.921	2	12
Political stability	1075	0.472	0.156	0	0.791
EVI	1218	39.356	10.483	16.382	70.049
Natural disasters index	1305	63.201	28.382	0	100
Exposure index	1262	39.597	9.786	21.993	66.881
Shock index	1259	39.196	17.009	2.373	88.173
Exchange rate	1167	6.351	4.763	1	15
ICT infrastructure	1000	0.237	0.065	0.18	0.58
Education	975	0.043	0.043	0.001	0.298
Innovation	525	0.014	0.033	0	0.142
Doing business	1000	0.324	0.121	0	0.827
Electric consumption	566	509.156	914.073	21.476	4766.652
Access to electricity urban	1092	60.631	24.65	3.5	100
Access to electricity rural	905	21.135	25.574	0.509	100

Table A12
Sources, and definitions of the data used.

Variables	Definition	Sources
1. Dependent variable		
Sustainable Development Index (SDI)	Measures the ecological efficiency of human development, recognizing that development must be achieved within planetary boundaries.	Hickel (2020)
2. Treatment variable		
Resource-backed loans (RBLs)	Takes value 1 if country i at period t has signed a resource-backed loans	Mihalyi et al. (2020)
Oil resource-backed loans	Takes value 1 if country i at period t has signed a loan on oil	Mihalyi et al. (2020)
Minerals resource-backed loans	Takes value 1 if country i at period t has signed a loan on minerals	Mihalyi et al. (2020)
Other resource-backed loans	Takes value 1 if country i at period t has signed a loan on tobacco and cocoa	Mihalyi et al. (2020)
3. Baseline model control variables or covariates		
Inflation rate	Inflation, average consumer prices (Percent change)	WDI
Natural resources rents	Natural resources rents to % GDP	WDI
ODA	Net ODA received (% of GNI)	WDI
FDI	Foreign direct investment, net inflows (% of GDP)	WDI
Resource depletion	Natural resource depletion is the sum of net forest depletion, energy depletion, and mineral depletion	WDI
4. Additional control variables		
Financial development	Continuous	IMF Financial Development database
Public Investment volatility	Standard deviation of the public investment estimated over 5-year moving window	World Economic Outlook (WEO)
Total investment	Total investment, percent of GDP	IMF-WEO
Resources-rich	Dummy	IMF dataset, Lledó et al. (2017)
Natural shock index	Continuous	Guillaumont (2009), Feindouno and Goujon (2016)
Economic vulnerability index (EVI)	Continuous	Guillaumont (2009), Feindouno and Goujon (2016)

(continued on next page)

Table A12 (continued)

Variables	Definition	Sources
Natural disasters index	Continuous	Guillaumont (2009), Feindouno and Goujon (2016)
Exposure index	Continuous	Guillaumont (2009), Feindouno and Goujon (2016)
Shock index	Continuous	Guillaumont (2009), Feindouno and Goujon (2016)
Rule of law	Continuous	Teorell et al. (2020)
Corruption	Continuous	Teorell et al. (2020)
Internal conflict	Continuous	ICRG
External conflict	Continuous	ICRG
Political stability	Index ranging from −2 to 2	WGI, Kaufmann et al. (2011)
Stock price volatility	Continuous	IMF-WEO
Public debt	Continuous	Kose et al. (2017)
Exchange rate regime	Dummy	Ilzetzki et al. (2017)
ICT infrastructure	Continuous	University of Notre Dame Global Adaptation Index
Education	Continuous	University of Notre Dame Global Adaptation Index
Innovation	Continuous	University of Notre Dame Global Adaptation Index
Doing business	Continuous	University of Notre Dame Global Adaptation Index
Electric consumption	Continuous	WDI
Access to electricity urban	Continuous	WDI
Access to electricity rural	Continuous	WDI
EITI countries	Dummy	World Bank Group.2016, (2016); Kinda and Thiombiano (2024)

Appendix C. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ecolecon.2024.108295>.

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